

ISC Class XII Mathematics

Chapter 12

Linear Programming

32 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered in this Chapter:

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Comprehensive Question Bank with Complete Solutions

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Formula & Standard Results Sheet

12.1 – Mathematical Formulation of LPPs

General LPP Structure

$$\text{Maximise/Minimise } Z = c_1x + c_2y$$

Subject to constraints $a_ix + b_iy \leq, \geq,$ or $= d_i$, and non-negativity (or minimum-quantity) restrictions on x, y .

Profit Per Unit

$$\text{Profit} = \text{Selling Price} - \text{Cost Price}$$

Used to convert selling/cost price data into the objective function's coefficients before formulating Z .

Consistent Units

$$1 \text{ hour} = 60 \text{ minutes}, \quad p\% \text{ of } x = \frac{p}{100}x$$

Always convert all given quantities (time, percentages) to the same unit before writing a constraint.

12.2 – Graphical Method: Corner Point Method

Corner Point Method (Bounded Region)

$$Z = ax + by$$

Evaluate Z at every corner point of the feasible region; the largest value is the maximum and the smallest is the minimum.

Corner Point Method (Unbounded Region)

$$ax + by < m \quad \text{or} \quad ax + by > M$$

Let m (or M) be the extreme value of Z found at the corner points. Test whether the open half-plane beyond it shares any point with the feasible region; if not, that value is the true optimum.

Multiple Optimal Solutions

$$P(\lambda) = \lambda P_1 + (1 - \lambda)P_2, \quad 0 \leq \lambda \leq 1$$

If Z takes the same optimal value at two corner points P_1, P_2 , it takes that same value at every point on the segment joining them.

Intercept Form of a Line

$$\frac{x}{a} + \frac{y}{b} = 1$$

Used to reverse-engineer a constraint's equation from a labelled feasible-region graph (intercepts a on the x -axis and b on the y -axis).

Parameter Relation from a Tied Optimum

$$Z(P_1) = Z(P_2)$$

When $Z = px + qy$ attains the same optimal value at two given corner points, equate Z at both points to derive the relation between the unknown parameters p and q .

12.1 – Mathematical Formulation of Linear Programming Problems

Question 1

Two tailors, A and B, earn Rs. 300 and Rs. 400 per day respectively. A can stitch 6 shirts and 4 pairs of trousers while B can stitch 10 shirts and 4 pairs of trousers per day. To find how many days should each of them work and if it is desired to produce at least 60 shirts and 32 pairs of trousers at a minimum labour cost, formulate this as an LPP.

TYPE 1 LPP Formulation from a Word Problem

Given: Let x be the number of days tailor A works and y be the number of days tailor B works. Daily wage of A = Rs. 300, daily wage of B = Rs. 400. A stitches 6 shirts and 4 trousers per day; B stitches 10 shirts and 4 trousers per day. Requirement: at least 60 shirts and at least 32 pairs of trousers.

To Find: Formulate the Linear Programming Problem (L.P.P.) to minimize the total labour cost.

Concept / Formula Used: Objective function: $Z = c_1x + c_2y$, subject to production requirement constraints $a_ix + b_iy \geq d_i$ and non-negativity $x \geq 0, y \geq 0$.

Solution:

Let x be the number of days tailor A works, and y be the number of days tailor B works.

Step 1: Formulate the Objective Function Tailor A earns Rs. 300 per day and tailor B earns Rs. 400 per day. The total labour cost Z to be minimized is:

$$\text{Minimize } Z = 300x + 400y$$

Step 2: Formulate the Constraints

- i. **Shirt Production Constraint:** Tailor A stitches 6 shirts per day and tailor B stitches 10 shirts per day. At least 60 shirts must be produced:

$$6x + 10y \geq 60$$

Dividing both sides by 2 to simplify:

$$3x + 5y \geq 30$$

- ii. **Trouser Production Constraint:** Both tailor A and tailor B stitch 4 pairs of trousers per day. At least 32 pairs of trousers must be produced:

$$4x + 4y \geq 32$$

Dividing both sides by 4 to simplify:

$$x + y \geq 8$$

iii. **Non-negativity Constraints:** The number of days worked cannot be negative:

$$x \geq 0, \quad y \geq 0$$

Final Answer

Minimize $Z = 300x + 400y$, subject to:

$$3x + 5y \geq 30, \quad x + y \geq 8, \quad x \geq 0, \quad y \geq 0$$

where x and y are the number of days tailor A and tailor B work, respectively.

Common Mistake: Forgetting to divide the raw shirt and trouser constraints ($6x + 10y \geq 60$ and $4x + 4y \geq 32$) down to their simplest form ($3x + 5y \geq 30$ and $x + y \geq 8$) does not make the formulation wrong, but it is the standard simplified form expected in board answers.

Question 2

A small firm manufactures necklaces and bracelets. The total number of necklaces and bracelets that they can handle per day is at most 24. It takes $\frac{1}{2}$ hour to make a necklace and 1 hour to make a bracelet. Maximum available time is 16 hours per day. If the profit is Rs. 100 per necklace and Rs. 300 per bracelet, formulate an L.P.P. to maximize profit, assuming that at least one of each item must be produced.

TYPE 1 LPP Formulation from a Word Problem

Given: Total daily production capacity ≤ 24 , time per necklace = $\frac{1}{2}$ hr, time per bracelet = 1 hr, maximum available time = 16 hrs, profit per necklace = Rs. 100, profit per bracelet = Rs. 300, minimum production ≥ 1 for each.

To Find: Formulate the Linear Programming Problem (L.P.P.) to maximize profit.

Concept / Formula Used: Maximize $Z = c_1x + c_2y$ subject to $a_1x + b_1y \leq c_1$, etc.

Solution:

Let x be the number of necklaces and y be the number of bracelets produced per day.

Step 1: Formulate the Objective Function Profit from one necklace is Rs. 100 and from one bracelet is Rs. 300. The total profit Z to be maximized is:

$$\text{Maximize } Z = 100x + 300y$$

Step 2: Formulate the Constraints

- i. **Total Production Capacity Constraint:** The total number of necklaces and bracelets handled per day is at most 24:

$$x + y \leq 24$$

- ii. **Time Constraint:** Time required to make x necklaces at $\frac{1}{2}$ hour each and y bracelets at 1 hour each cannot exceed 16 hours:

$$\frac{1}{2}x + y \leq 16 \implies x + 2y \leq 32$$

- iii. **Minimum Production Constraints:** At least one of each item must be produced:

$$x \geq 1, \quad y \geq 1$$

Final Answer

Maximize $Z = 100x + 300y$, subject to:

$$x + y \leq 24, \quad x + 2y \leq 32, \quad x \geq 1, \quad y \geq 1$$

where x and y are the number of necklaces and bracelets produced daily.

Common Mistake: Forgetting to include the minimum production constraint $x \geq 1, y \geq 1$ and defaulting to $x \geq 0, y \geq 0$ will lead to an incorrect formulation.

Question 3

A furniture dealer deals in only two items—tables and chairs. He has Rs. 20000 to invest and a space to store atmost 80 pieces. A table costs him Rs. 800 and a chair costs him Rs. 200. He can sell a table for Rs. 950 and a chair for Rs. 280. Assume that he can sell all the items that he buys. Formulate this problem as an L.P.P. so that he can maximize his profit.

TYPE 1 LPP Formulation from a Word Problem

Given: Investment limit = Rs. 20000, maximum storage capacity = 80 pieces, cost price of a table = Rs. 800, cost price of a chair = Rs. 200, selling price of a table = Rs. 950, selling price of a chair = Rs. 280.

To Find: Formulate the Linear Programming Problem (L.P.P.) to maximize the total profit.

Concept / Formula Used: Profit per unit = Selling Price – Cost Price;
Total Profit $Z = P_1x + P_2y$, subject to linear inequalities representing capacity and budget constraints.

Solution:

Let x be the number of tables purchased and sold, and y be the number of chairs purchased and sold.

Step 1: Calculate profit per item

$$\begin{aligned}\text{Profit on one table} &= \text{Selling price of a table} - \text{Cost price of a table} \\ &= 950 - 800 \\ &= \text{Rs. } 150\end{aligned}$$

$$\begin{aligned}\text{Profit on one chair} &= \text{Selling price of a chair} - \text{Cost price of a chair} \\ &= 280 - 200 \\ &= \text{Rs. } 80\end{aligned}$$

Step 2: Formulate the Objective Function The dealer wants to maximize total profit Z from selling x tables and y chairs:

$$\text{Maximize } Z = 150x + 80y$$

Step 3: Formulate the Constraints

- i. **Investment Constraint:** The total cost of x tables and y chairs cannot exceed the available capital of Rs. 20000:

$$800x + 200y \leq 20000$$

Dividing both sides by 200:

$$4x + y \leq 100$$

- ii. **Storage Capacity Constraint:** The dealer can store at most 80 pieces in total:

$$x + y \leq 80$$

- iii. **Non-negativity Constraints:** The number of tables and chairs cannot be negative:

$$x \geq 0, \quad y \geq 0$$

Step 4: Mathematical Formulation of L.P.P. Combining the objective function and constraints:

$$\text{Maximize } Z = 150x + 80y$$

subject to the constraints:

$$\begin{aligned}4x + y &\leq 100 \\ x + y &\leq 80 \\ x \geq 0, \quad y &\geq 0\end{aligned}$$

Final Answer

Maximize $Z = 150x + 80y$, subject to the constraints:

$$4x + y \leq 100, \quad x + y \leq 80, \quad x \geq 0, \quad y \geq 0$$

where x and y represent the number of tables and chairs, respectively.

Common Mistake: A common mistake is using the selling prices (Rs. 950 and Rs. 280) directly in the objective function instead of calculating the net profit per item (Selling Price – Cost Price).

Question 4

Each unit of Product A that a company produces, is sold for Rs. 100 with a production cost of Rs. 60 and each unit of Product B is sold for Rs. 150 with production cost of Rs. 90. On a given day, the company has a budget of Rs. 8000 to spend on production. The production process makes it such that they can only produce a maximum of 100 units each day. Also, the number of Product B produced cannot be more than twice as many of Product A.

Frame a linear programming problem to determine how many units of Product A and B should the company produce in a day in order to maximize their profit?

(Note: No need to find the feasible region and optimal solution.)

TYPE 1 LPP Formulation from a Word Problem

Given: Let x be the number of units of Product A produced per day, and y be the number of units of Product B produced per day.

- Selling price of Product A = Rs. 100 per unit, Production cost = Rs. 60 per unit
- Selling price of Product B = Rs. 150 per unit, Production cost = Rs. 90 per unit
- Daily production budget = Rs. 8000
- Maximum total units produced per day = 100
- Maximum constraint on Product B = at most twice the units of Product A

Concept / Formula Used: Profit per unit = Selling Price – Production Cost
A Linear Programming Problem (L.P.P.) consists of an objective function to be maximized or minimized subject to a set of linear inequality constraints and non-negativity restrictions.

Solution:

Step 1: Define the decision variables. Let x denote the number of units of Product A produced per day. Let y denote the number of units of Product B produced per day.

Step 2: Determine the profit per unit and formulate the objective function.

$$\text{Profit per unit of Product A} = \text{Rs. } 100 - \text{Rs. } 60 = \text{Rs. } 40$$

$$\text{Profit per unit of Product B} = \text{Rs. } 150 - \text{Rs. } 90 = \text{Rs. } 60$$

Thus, the total daily profit P obtained by producing x units of A and y units of B is given by:

$$P = 40x + 60y$$

The objective of the company is to maximize total profit P .

Step 3: Formulate the constraints.

- (i) **Investment / Budget Constraint:** The production cost of x units of Product A and y units of Product B is $60x + 90y$. Since the available daily budget is Rs. 8000:

$$60x + 90y \leq 8000$$

Dividing by 10, this simplifies to:

$$6x + 9y \leq 800$$

- (ii) **Maximum Production Constraint:** The total number of units produced per day cannot exceed 100:

$$x + y \leq 100$$

- (iii) **Product B Production Limit Constraint:** The number of units of Product B produced cannot be more than twice the number of units of Product A:

$$\begin{aligned} y &\leq 2x \\ \Rightarrow 2x - y &\geq 0 \end{aligned}$$

- (iv) **Non-negativity Constraints:** The number of units produced cannot be negative:

$$x \geq 0, \quad y \geq 0$$

Final Answer

Maximize profit:

$$P = 40x + 60y$$

Subject to the constraints:

$$60x + 90y \leq 8000 \quad (\text{i.e., } 6x + 9y \leq 800)$$

$$x + y \leq 100$$

$$y \leq 2x \quad (\text{i.e., } 2x - y \geq 0)$$

$$x \geq 0, \quad y \geq 0$$

where x and y represent the number of units of Product A and Product B produced daily, respectively.

Common Mistake: Confusing the direction of the relative constraint "Product B cannot be more than twice Product A" as $x \leq 2y$ instead of $y \leq 2x$. Always state the dependent quantity (y) on the left-hand side first to verify.

Question 5

A packaging company has the capacity to produce rectangular boxes (r) and circular boxes (c). Each rectangular box takes 2 minutes to make and it sells for a profit of Rs. 4. Each circular box takes 3 minutes to make and it sells for a profit of Rs. 5. Their client requests for 25 boxes to be ready in one hour. The packaging company

wants to maximise their profit from this order.
Frame this optimisation problem as a linear programming problem.

TYPE 1 LPP Formulation from a Word Problem

Given: Let r denote the number of rectangular boxes and c denote the number of circular boxes.

Profit per rectangular box = Rs. 4

Profit per circular box = Rs. 5

Time taken per rectangular box = 2 minutes

Time taken per circular box = 3 minutes

Total available time = 1 hour = 60 minutes

Total boxes requested by client = 25 boxes

Concept / Formula Used: To formulate a Linear Programming Problem (LPP):

Maximise $Z = c_1r + c_2c$

subject to constraints: $a_1r + b_1c \leq T$, $r + c = N$, and non-negativity $r \geq 0, c \geq 0$.

Solution:

Step 1: Identify the Decision Variables Let r be the number of rectangular boxes produced. Let c be the number of circular boxes produced.

Step 2: Formulate the Objective Function Profit earned from 1 rectangular box = Rs. 4 Profit earned from 1 circular box = Rs. 5 Total profit Z from producing r rectangular boxes and c circular boxes is:

$$Z = 4r + 5c$$

Since the objective is to maximize profit, the objective function is:

$$\text{Maximise } Z = 4r + 5c$$

Step 3: Formulate the Time Constraint Time taken to make 1 rectangular box = 2 minutes. Time taken to make 1 circular box = 3 minutes. Total time required to produce r rectangular boxes and c circular boxes is:

$$\text{Total Time} = 2r + 3c \text{ minutes}$$

The total time available is 1 hour = 60 minutes. Therefore, the time constraint is:

$$2r + 3c \leq 60$$

Step 4: Formulate the Order/Demand Constraint The client requests exactly 25 boxes to be ready in one hour. Therefore, the total number of boxes produced must equal 25:

$$r + c = 25$$

Step 5: Formulate the Non-Negativity Constraints Since the number of boxes produced cannot be negative:

$$r \geq 0, \quad c \geq 0$$

Step 6: Complete LPP Statement Combining the objective function and all constraints gives the complete linear programming formulation:

$$\begin{aligned} &\text{Maximise } Z = 4r + 5c \\ &\text{subject to constraints:} \\ &\quad r + c = 25 \\ &\quad 2r + 3c \leq 60 \\ &\quad r \geq 0, \quad c \geq 0 \end{aligned}$$

Final Answer

The linear programming problem is formulated as follows:

$$\text{Maximise } Z = 4r + 5c$$

$$\text{subject to constraints: } r + c = 25$$

$$2r + 3c \leq 60$$

$$r \geq 0, \quad c \geq 0$$

where r = number of rectangular boxes and c = number of circular boxes.

Common Mistake: Forgetting to convert 1 hour to 60 minutes before writing the time constraint is a common error, leading to an incorrect constraint such as $2r + 3c \leq 1$. Always ensure all time units are uniform throughout the problem.

12.2 – Graphical Method: Solving LPPs by the Corner Point Method

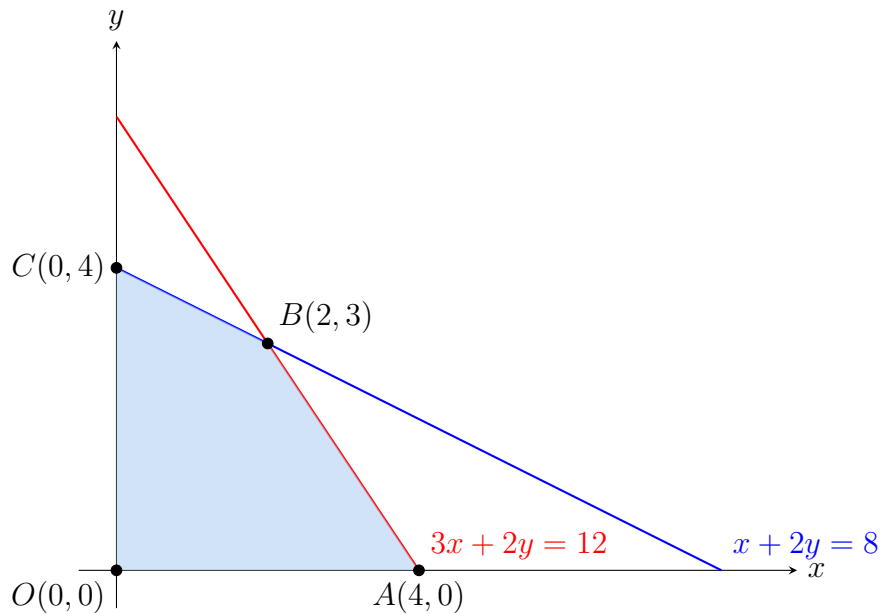
Question 1

Minimize $Z = -3x + 4y$ subject to the constraints:

$$x + 2y \leq 8, \quad 3x + 2y \leq 12, \quad x \geq 0, \quad y \geq 0$$

TYPE 2 Corner Point Method – Evaluate Z at Vertices for Optimum

★ Likely Exam Question



Feasible region for Question 1

Given: $Z = -3x + 4y$ subject to $x + 2y \leq 8$, $3x + 2y \leq 12$, $x \geq 0$, $y \geq 0$.

To Find: The minimum value of Z and the point at which it occurs.

Concept / Formula Used: Evaluate the objective function $Z = ax + by$ at each corner point of the feasible region. The smallest value represents the minimum when the region is bounded.

Solution:

First, convert the inequations into equations to find the boundary lines:

$$L_1 : x + 2y = 8$$

$$L_2 : 3x + 2y = 12$$

To find the intersection point of line L_1 and line L_2 : Subtract equation L_1 from equation L_2 :

$$\begin{aligned} (3x + 2y) - (x + 2y) &= 12 - 8 \\ 2x &= 4 \\ x &= 2 \end{aligned}$$

Substitute $x = 2$ into L_1 :

$$\begin{aligned} 2 + 2y &= 8 \\ 2y &= 6 \\ y &= 3 \end{aligned}$$

Thus, the intersection point is $B(2, 3)$.

The boundary points on the axes are: For L_1 : when $y = 0$, $x = 8$; when $x = 0$, $y = 4 \implies (0, 4)$. For L_2 : when $y = 0$, $x = 4 \implies (4, 0)$; when $x = 0$, $y = 6$.

Since $x \geq 0$ and $y \geq 0$, the feasible region is in the first quadrant and bounded by the corner points:

$$O(0, 0), \quad A(4, 0), \quad B(2, 3), \quad C(0, 4)$$

Now evaluate the objective function $Z = -3x + 4y$ at each corner point:

$$\text{At } O(0, 0): \quad Z = -3(0) + 4(0) = 0$$

$$\text{At } A(4, 0): \quad Z = -3(4) + 4(0) = -12$$

$$\text{At } B(2, 3): \quad Z = -3(2) + 4(3) = -6 + 12 = 6$$

$$\text{At } C(0, 4): \quad Z = -3(0) + 4(4) = 16$$

Comparing these values, the minimum value of Z is -12 , which occurs at $(4, 0)$.

Final Answer

Minimum value of $Z = -12$ at $x = 4, y = 0$.

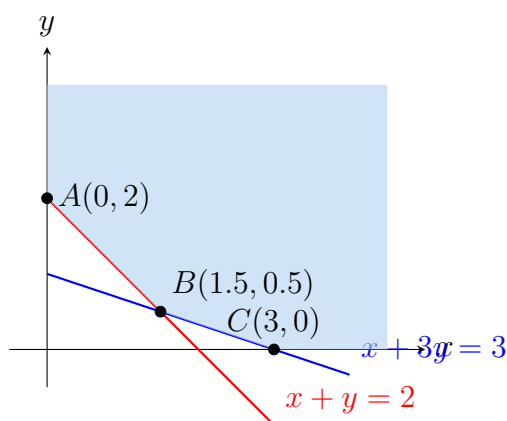
Common Mistake: Carefully evaluate the negative signs in the objective function $Z = -3x + 4y$; confusing $-3(4)$ with $+12$ would incorrectly point to $(0, 0)$ as the minimum.

Question 2

Minimize $Z = 3x + 5y$ subject to the constraints:

$$x + 3y \geq 3, \quad x + y \geq 2, \quad x \geq 0, \quad y \geq 0$$

TYPE 2 Corner Point Method – Evaluate Z at Vertices for Optimum



Unbounded feasible region for Question 2

Given: $Z = 3x + 5y$ subject to $x + 3y \geq 3$, $x + y \geq 2$, $x \geq 0, y \geq 0$.

To Find: The minimum value of Z .

Concept / Formula Used: For unbounded feasible regions, let m be the minimum value of the objective function $Z = ax + by$ at the corner points. Check if the open half-plane $ax + by < m$ has any points in common with the feasible region. If no points are in common, m is the true minimum.

Solution:

Convert constraints to equations to find boundary lines:

$$L_1 : x + 3y = 3$$

$$L_2 : x + y = 2$$

Find the intersection of L_1 and L_2 : Subtract L_2 from L_1 :

$$(x + 3y) - (x + y) = 3 - 2$$

$$2y = 1$$

$$y = \frac{1}{2} = 0.5$$

Substitute $y = 0.5$ into L_2 :

$$x + 0.5 = 2$$

$$x = 1.5 = \frac{3}{2}$$

So the intersection point is $B\left(\frac{3}{2}, \frac{1}{2}\right)$.

The boundary points on the axes are: For L_1 : $(3, 0)$ and $(0, 1)$. For L_2 : $(2, 0)$ and $(0, 2)$.

Since the inequalities are $\geq$, the feasible region is unbounded and lies above the lines, with corner points:

$$A(0, 2), \quad B\left(\frac{3}{2}, \frac{1}{2}\right), \quad C(3, 0)$$

Evaluate $Z = 3x + 5y$ at these corner points:

$$\text{At } A(0, 2) : Z = 3(0) + 5(2) = 10$$

$$\text{At } B\left(\frac{3}{2}, \frac{1}{2}\right) : Z = 3\left(\frac{3}{2}\right) + 5\left(\frac{1}{2}\right) = \frac{9}{2} + \frac{5}{2} = \frac{14}{2} = 7$$

$$\text{At } C(3, 0) : Z = 3(3) + 5(0) = 9$$

Smallest value of Z at the corner points is 7. Since the region is unbounded, check if the open half-plane $3x + 5y < 7$ shares points with the feasible region. The line $3x + 5y = 7$ passes through $(0, 1.4)$ and $(2.33, 0)$ and point $B(1.5, 0.5)$. Testing origin $(0, 0)$: $3(0) + 5(0) = 0 < 7$, which lies below the feasible region. Thus, the open half-plane $3x + 5y < 7$ has no common points with the feasible region.

Final Answer

Minimum value of $Z = 7$ at $x = \frac{3}{2}, y = \frac{1}{2}$.

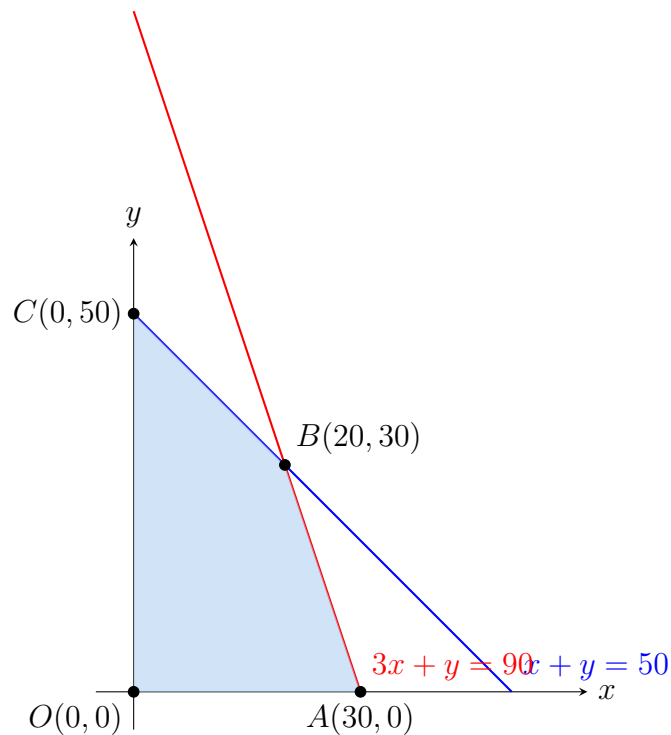
Common Mistake: For unbounded regions, forgetting to verify whether $3x + 5y < 7$ intersects the feasible region can lose marks in ISC board exams.

Question 3

Maximize $Z = 4x + y$, subject to the constraints:

$$x + y \leq 50, \quad 3x + y \leq 90, \quad x \geq 0, \quad y \geq 0$$

TYPE 2 Corner Point Method – Evaluate Z at Vertices for Optimum



Feasible region for Question 3

Given: $Z = 4x + y$ subject to $x + y \leq 50$, $3x + y \leq 90$, $x \geq 0$, $y \geq 0$.

To Find: The maximum value of Z .

Solution:

Convert the given constraints into equations:

$$L_1 : x + y = 50$$

$$L_2 : 3x + y = 90$$

Find the point of intersection of L_1 and L_2 : Subtract L_1 from L_2 :

$$(3x + y) - (x + y) = 90 - 50$$

$$2x = 40$$

$$x = 20$$

Substitute $x = 20$ into L_1 :

$$\begin{aligned} 20 + y &= 50 \\ y &= 30 \end{aligned}$$

The point of intersection is $B(20, 30)$.

Find axis intercepts: For L_1 : $(50, 0)$ and $(0, 50)$. For L_2 : $(30, 0)$ and $(0, 90)$.

The bounded feasible region is defined by the corner points:

$$O(0, 0), \quad A(30, 0), \quad B(20, 30), \quad C(0, 50)$$

Evaluate $Z = 4x + y$ at these corner points:

$$\begin{aligned} \text{At } O(0, 0) : \quad Z &= 4(0) + 0 = 0 \\ \text{At } A(30, 0) : \quad Z &= 4(30) + 0 = 120 \\ \text{At } B(20, 30) : \quad Z &= 4(20) + 30 = 80 + 30 = 110 \\ \text{At } C(0, 50) : \quad Z &= 4(0) + 50 = 50 \end{aligned}$$

Comparing values, the maximum value of Z is 120.

Final Answer

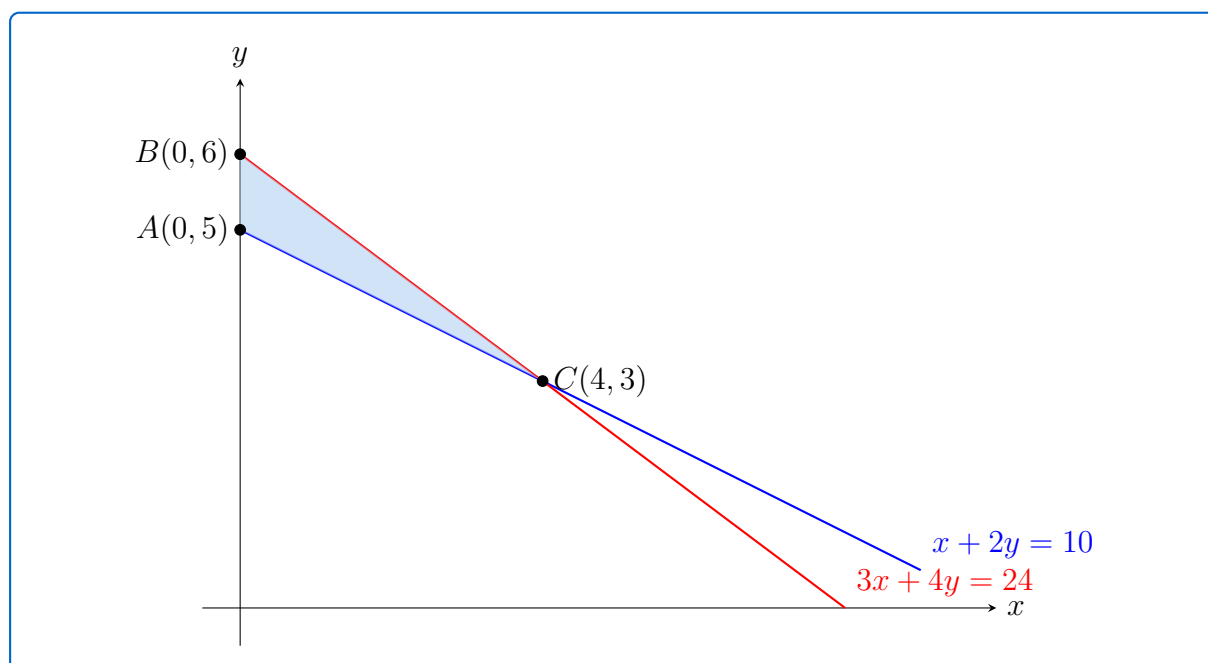
Maximum value of $Z = 120$ at $x = 30, y = 0$.

Question 4

Minimize $Z = 200x + 500y$, subject to the constraints:

$$x + 2y \geq 10, \quad 3x + 4y \leq 24, \quad x \geq 0, \quad y \geq 0$$

TYPE 2 Corner Point Method – Evaluate Z at Vertices for Optimum



Feasible region for Question 4

Given: $Z = 200x + 500y$ subject to $x + 2y \geq 10$, $3x + 4y \leq 24$, $x \geq 0$, $y \geq 0$.

To Find: The minimum value of Z .

Solution:

Convert the inequalities to boundary line equations:

$$L_1 : x + 2y = 10$$

$$L_2 : 3x + 4y = 24$$

Find the intersection of L_1 and L_2 : Multiply L_1 by 2:

$$2x + 4y = 20$$

Subtract this from L_2 :

$$\begin{aligned}(3x + 4y) - (2x + 4y) &= 24 - 20 \\ x &= 4\end{aligned}$$

Substitute $x = 4$ into L_1 :

$$4 + 2y = 10$$

$$2y = 6$$

$$y = 3$$

Intersection point is $C(4, 3)$.

Determine y -axis intercepts: For L_1 : $x = 0 \implies y = 5 \implies (0, 5)$. For L_2 : $x = 0 \implies y = 6 \implies (0, 6)$.

The region satisfying $x + 2y \geq 10$ (above L_1) and $3x + 4y \leq 24$ (below L_2) with $x \geq 0$, $y \geq 0$ forms a triangle with corner points:

$$A(0, 5), \quad B(0, 6), \quad C(4, 3)$$

Evaluate $Z = 200x + 500y$ at each corner point:

$$\text{At } A(0, 5) : Z = 200(0) + 500(5) = 2500$$

$$\text{At } B(0, 6) : Z = 200(0) + 500(6) = 3000$$

$$\text{At } C(4, 3) : Z = 200(4) + 500(3) = 800 + 1500 = 2300$$

Comparing the values, the minimum value of Z is 2300.

Final Answer

Minimum value of $Z = 2300$ at $x = 4, y = 3$.

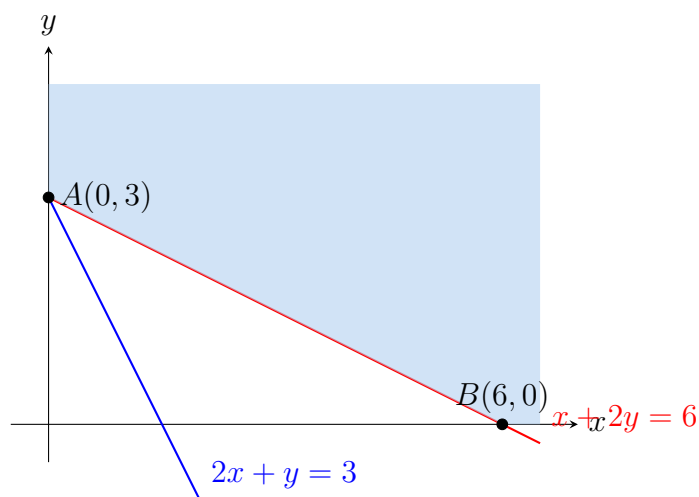
Question 5

Minimize $Z = x + 2y$ subject to the constraints:

$$2x + y \geq 3, \quad x + 2y \geq 6, \quad x \geq 0, \quad y \geq 0$$

Show that the minimum value of Z occurs at more than two points.

TYPE 3 Corner Point Method – Multiple Optimal Solutions (Line Segment)



Feasible region for Question 5 showing parallel objective line

To Prove: Show that the minimum value of $Z = x + 2y$ occurs at more than two points.

Concept / Formula Used: If the objective function $Z = ax + by$ yields the same optimal value at two distinct corner points, then it yields that same optimal value at every point on the line segment joining them.

Solution:

Convert the inequations into boundary equations:

$$L_1 : 2x + y = 3$$

$$L_2 : x + 2y = 6$$

Find intersection of L_1 and L_2 : Multiply L_1 by 2:

$$4x + 2y = 6$$

Subtract L_2 from this equation:

$$\begin{aligned} (4x + 2y) - (x + 2y) &= 6 - 6 \\ 3x &= 0 \implies x = 0 \end{aligned}$$

Substitute $x = 0$ into L_2 :

$$0 + 2y = 6 \implies y = 3$$

So the intersection point is $A(0, 3)$.

For line L_2 , when $y = 0$, $x = 6 \implies B(6, 0)$. The corner points of the unbounded feasible region defined by $2x + y \geq 3$ and $x + 2y \geq 6$ with $x \geq 0, y \geq 0$ are:

$$A(0, 3) \quad \text{and} \quad B(6, 0)$$

Evaluate $Z = x + 2y$ at these corner points:

$$\text{At } A(0, 3) : Z = 0 + 2(3) = 6$$

$$\text{At } B(6, 0) : Z = 6 + 2(0) = 6$$

Both corner points yield the same minimum value of 6. Furthermore, the objective function $Z = x + 2y$ has the same slope as the constraint line $x + 2y = 6$. Therefore, for any point (x, y) lying on the line segment joining $A(0, 3)$ and $B(6, 0)$, the value of Z is:

$$Z = x + 2y = 6$$

Since there are infinitely many points on the line segment joining $(0, 3)$ and $(6, 0)$, the minimum value of Z occurs at infinitely many points (which is more than two points).

Hence Proved

Minimum value of $Z = 6$ occurs at all points on the line segment joining $(0, 3)$ and $(6, 0)$, hence occurring at more than two points.

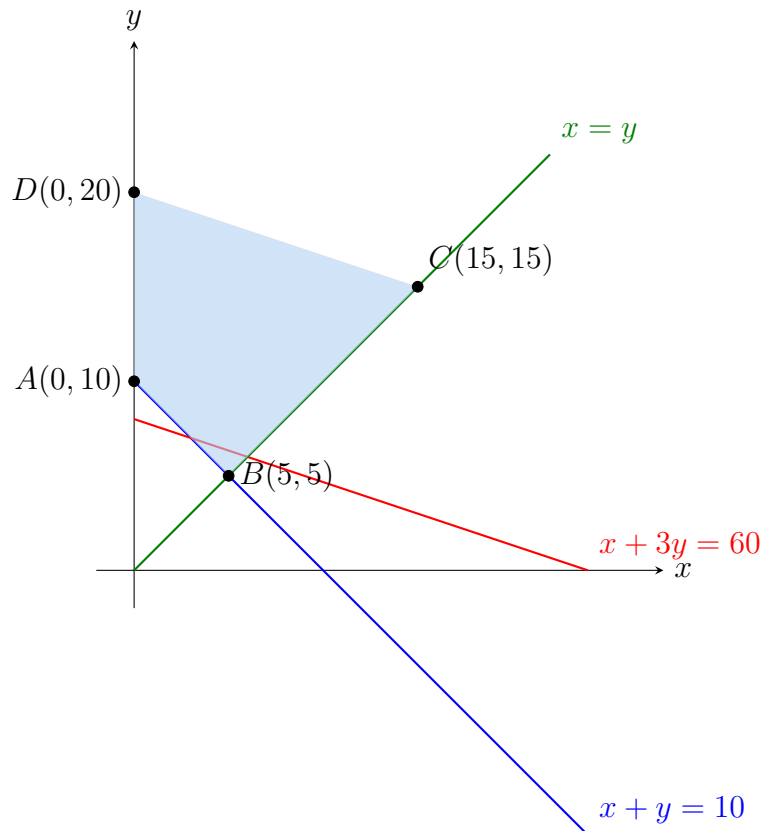
Question 6

Minimize and maximize $Z = 3x + 9y$ subject to the constraints:

$$x + y \geq 10, \quad x + 3y \leq 60, \quad x \leq y, \quad x \geq 0, \quad y \geq 0$$

TYPE 2 Corner Point Method – Evaluate Z at Vertices for Optimum

★ Likely Exam Question



Feasible region for Question 6

Given: $Z = 3x + 9y$ subject to $x + y \geq 10$, $x + 3y \leq 60$, $x \leq y$, $x \geq 0$, $y \geq 0$.

To Find: The minimum and maximum values of Z .

Solution:

Convert constraints to boundary equations:

$$L_1 : x + y = 10$$

$$L_2 : x + 3y = 60$$

$$L_3 : x = y$$

Find intersection points: 1. Intersection of L_1 and L_3 : Substitute $y = x$ into $x + y = 10$:

$$2x = 10 \implies x = 5, \quad y = 5 \implies B(5, 5)$$

2. Intersection of L_2 and L_3 : Substitute $y = x$ into $x + 3y = 60$:

$$4x = 60 \implies x = 15, \quad y = 15 \implies C(15, 15)$$

3. y -axis intercepts ($x = 0$): For L_1 : $x = 0 \implies y = 10 \implies A(0, 10)$. For L_2 : $x = 0 \implies y = 20 \implies D(0, 20)$.

The bounded feasible region has corner points:

$$A(0, 10), \quad B(5, 5), \quad C(15, 15), \quad D(0, 20)$$

Evaluate $Z = 3x + 9y$ at each corner point:

$$\text{At } A(0, 10) : Z = 3(0) + 9(10) = 90$$

$$\text{At } B(5, 5) : Z = 3(5) + 9(5) = 15 + 45 = 60$$

$$\text{At } C(15, 15) : Z = 3(15) + 9(15) = 45 + 135 = 180$$

$$\text{At } D(0, 20) : Z = 3(0) + 9(20) = 180$$

Comparison yields: Minimum value of Z is 60, occurring at $(5, 5)$. Maximum value of Z is 180, occurring at both $(15, 15)$ and $(0, 20)$ (and all points on the segment joining them, as $Z = 3(x + 3y)$ is parallel to $x + 3y = 60$).

Final Answer

Minimum value of $Z = 60$ at $x = 5, y = 5$; Maximum value of $Z = 180$ at all points on the line segment joining $(15, 15)$ and $(0, 20)$.

Question 7 (i)

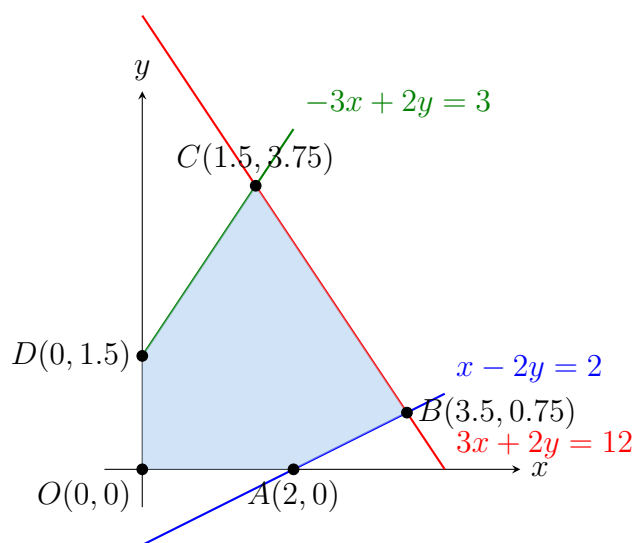
Solve the following Linear Programming problem graphically:

Maximise $Z = 5x + 2y$ subject to:

$$x - 2y \leq 2, \quad 3x + 2y \leq 12, \quad -3x + 2y \leq 3, \quad x \geq 0, \quad y \geq 0$$

TYPE 2 Corner Point Method – Evaluate Z at Vertices for Optimum

★ Likely Exam Question



Feasible region for Question 7(i)

Given: $Z = 5x + 2y$ subject to $x - 2y \leq 2$, $3x + 2y \leq 12$, $-3x + 2y \leq 3$, $x \geq 0$, $y \geq 0$.

To Find: The maximum value of Z .

Solution:

Convert inequations to equations:

$$L_1 : x - 2y = 2$$

$$L_2 : 3x + 2y = 12$$

$$L_3 : -3x + 2y = 3$$

Find corner points of the feasible region: 1. Intersection of L_1 and L_2 : Add L_1 and L_2 :

$$\begin{aligned}(x - 2y) + (3x + 2y) &= 2 + 12 \\ 4x &= 14 \implies x = 3.5 = \frac{7}{2}\end{aligned}$$

Substitute into L_1 :

$$3.5 - 2y = 2 \implies 2y = 1.5 \implies y = 0.75 = \frac{3}{4} \implies B\left(\frac{7}{2}, \frac{3}{4}\right)$$

2. Intersection of L_2 and L_3 : Add L_2 and L_3 :

$$\begin{aligned}(3x + 2y) + (-3x + 2y) &= 12 + 3 \\ 4y &= 15 \implies y = 3.75 = \frac{15}{4}\end{aligned}$$

Substitute into L_2 :

$$3x + 2(3.75) = 12 \implies 3x = 4.5 \implies x = 1.5 = \frac{3}{2} \implies C\left(\frac{3}{2}, \frac{15}{4}\right)$$

3. Axis intercepts: For L_1 on $y = 0$: $x = 2 \implies A(2, 0)$. For L_3 on $x = 0$: $y = 1.5 \implies D\left(0, \frac{3}{2}\right)$. Origin: $O(0, 0)$.

Feasible region corner points are $O(0, 0)$, $A(2, 0)$, $B(3.5, 0.75)$, $C(1.5, 3.75)$, $D(0, 1.5)$.

Evaluate $Z = 5x + 2y$:

$$\text{At } O(0, 0) : Z = 0$$

$$\text{At } A(2, 0) : Z = 5(2) + 2(0) = 10$$

$$\text{At } B(3.5, 0.75) : Z = 5(3.5) + 2(0.75) = 17.5 + 1.5 = 19$$

$$\text{At } C(1.5, 3.75) : Z = 5(1.5) + 2(3.75) = 7.5 + 7.5 = 15$$

$$\text{At } D(0, 1.5) : Z = 5(0) + 2(1.5) = 3$$

Maximum value of Z is 19.

Final Answer

Maximum value of $Z = 19$ at $x = \frac{7}{2}$, $y = \frac{3}{4}$.

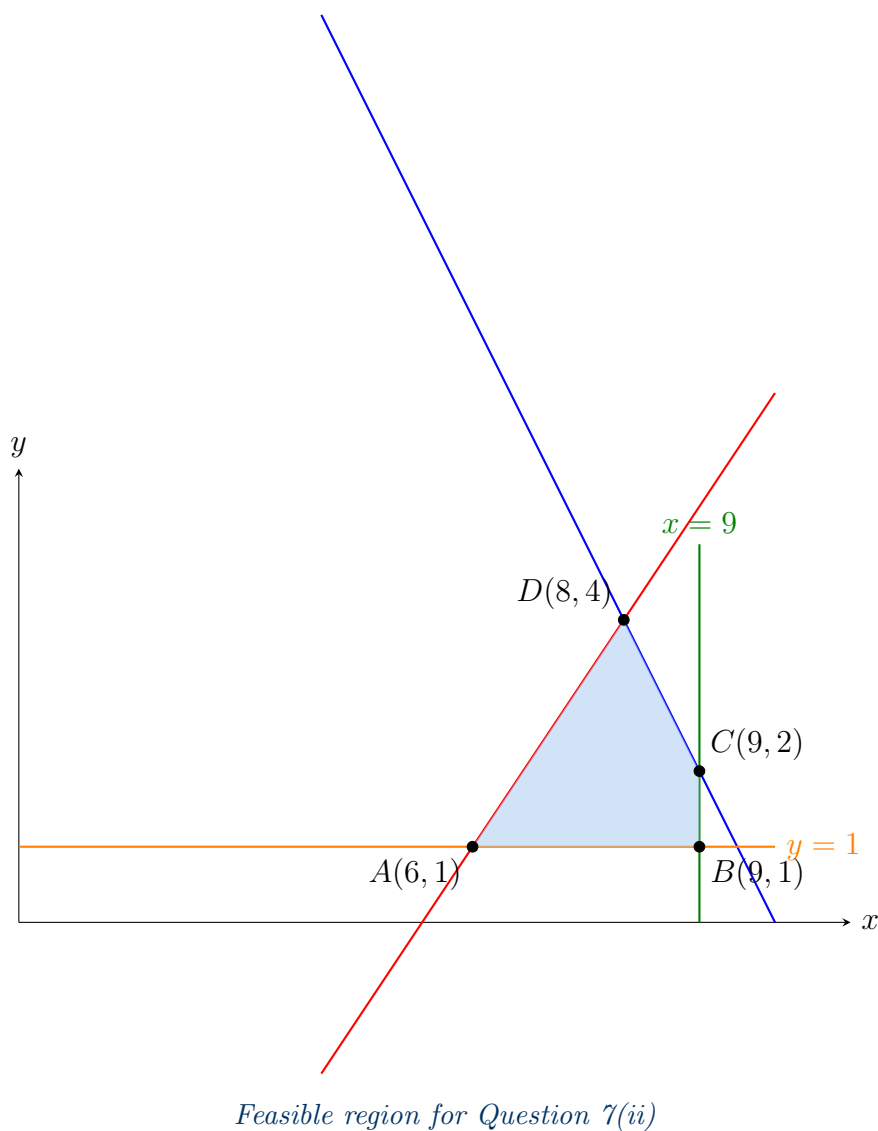
Question 7 (ii)

Maximise $Z = 4x - 3y$, subject to the following constraints:

$$2x + y \leq 20, \quad 3x - 2y \geq 16, \quad x \leq 16, \quad x \leq 9, \quad y \geq 1$$

Also, find Z 's maximum value.

TYPE 2 Corner Point Method – Evaluate Z at Vertices for Optimum



Given: $Z = 4x - 3y$ subject to $2x + y \leq 20$, $3x - 2y \geq 16$, $x \leq 9$, $y \geq 1$.

To Find: The maximum value of Z .

Solution:

Notice that $x \leq 16$ is redundant since $x \leq 9$. Convert constraints to boundary lines:

$$\begin{aligned} L_1 : 2x + y &= 20 \\ L_2 : 3x - 2y &= 16 \\ L_3 : x &= 9 \\ L_4 : y &= 1 \end{aligned}$$

Determine corner points of the feasible region by intersection of boundary lines: 1. Intersection of L_4 ($y = 1$) and L_2 :

$$3x - 2(1) = 16 \implies 3x = 18 \implies x = 6 \implies A(6, 1)$$

2. Intersection of L_4 ($y = 1$) and L_3 ($x = 9$):

$$B(9, 1)$$

3. Intersection of L_3 ($x = 9$) and L_1 :

$$2(9) + y = 20 \implies y = 2 \implies C(9, 2)$$

4. Intersection of L_1 and L_2 : Multiply L_1 by 2:

$$4x + 2y = 40$$

Add to L_2 :

$$\begin{aligned} (4x + 2y) + (3x - 2y) &= 40 + 16 \\ 7x &= 56 \implies x = 8 \end{aligned}$$

Substitute $x = 8$ into L_1 :

$$2(8) + y = 20 \implies y = 4 \implies D(8, 4)$$

Feasible region corner points are $A(6, 1)$, $B(9, 1)$, $C(9, 2)$, $D(8, 4)$.

Evaluate $Z = 4x - 3y$ at each corner point:

$$\begin{aligned} \text{At } A(6, 1) : Z &= 4(6) - 3(1) = 24 - 3 = 21 \\ \text{At } B(9, 1) : Z &= 4(9) - 3(1) = 36 - 3 = 33 \\ \text{At } C(9, 2) : Z &= 4(9) - 3(2) = 36 - 6 = 30 \\ \text{At } D(8, 4) : Z &= 4(8) - 3(4) = 32 - 12 = 20 \end{aligned}$$

The maximum value of Z is 33.

Final Answer

Maximum value of $Z = 33$ at $x = 9, y = 1$.

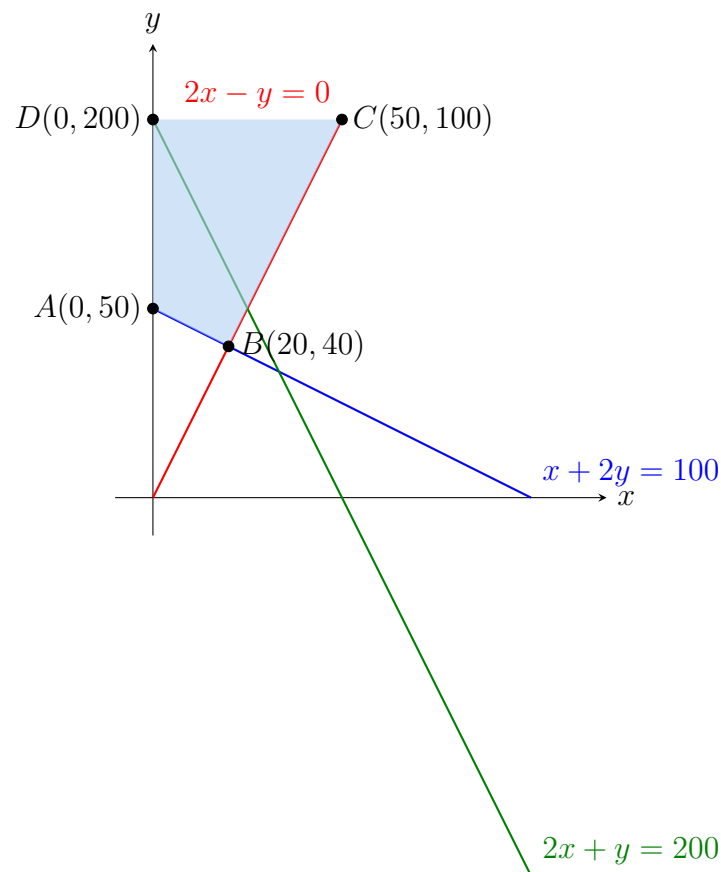
Question 8

Minimize and maximize $Z = x + 2y$ subject to the constraints:

$$x + 2y \geq 100, \quad 2x - y \leq 0, \quad 2x + y \leq 200, \quad x \geq 0, \quad y \geq 0$$

TYPE 2 Corner Point Method – Evaluate Z at Vertices for Optimum

★ Likely Exam Question



Feasible region for Question 8

Given: $Z = x + 2y$ subject to $x + 2y \geq 100$, $2x - y \leq 0$, $2x + y \leq 200$, $x \geq 0$, $y \geq 0$.

To Find: The minimum and maximum values of Z .

Solution:

Convert constraints to boundary line equations:

$$L_1 : x + 2y = 100$$

$$L_2 : 2x - y = 0 \implies y = 2x$$

$$L_3 : 2x + y = 200$$

Determine the intersection points of boundary lines: 1. Intersection of L_1 and L_2 : Substitute $y = 2x$ into L_1 :

$$x + 2(2x) = 100 \implies 5x = 100 \implies x = 20, \quad y = 40 \implies B(20, 40)$$

2. Intersection of L_2 and L_3 : Substitute $y = 2x$ into L_3 :

$$2x + 2x = 200 \implies 4x = 200 \implies x = 50, \quad y = 100 \implies C(50, 100)$$

3. Intercepts on y -axis ($x = 0$): For L_1 : $x = 0 \implies y = 50 \implies A(0, 50)$. For L_3 : $x = 0 \implies y = 200 \implies D(0, 200)$.

The feasible region is a quadrilateral with corner points $A(0, 50)$, $B(20, 40)$, $C(50, 100)$, $D(0, 200)$.

Evaluate $Z = x + 2y$ at each corner point:

$$\text{At } A(0, 50) : Z = 0 + 2(50) = 100$$

$$\text{At } B(20, 40) : Z = 20 + 2(40) = 100$$

$$\text{At } C(50, 100) : Z = 50 + 2(100) = 250$$

$$\text{At } D(0, 200) : Z = 0 + 2(200) = 400$$

From the evaluated values: Minimum value of $Z = 100$ occurs at both $(0, 50)$ and $(20, 40)$ (and all points on the line segment joining them). Maximum value of $Z = 400$ occurs at $(0, 200)$.

Final Answer

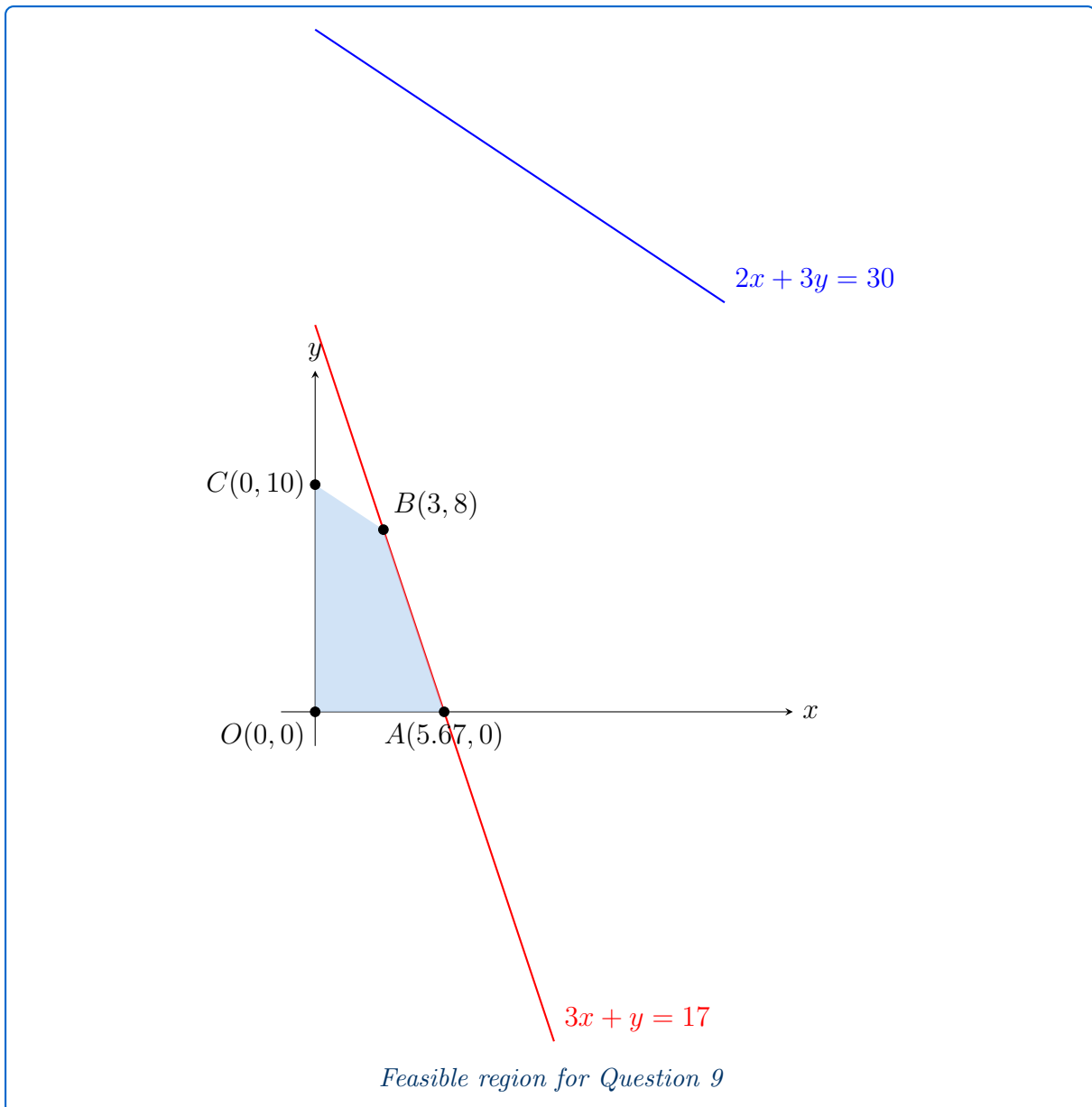
Minimum value of $Z = 100$ at all points on the segment joining $(0, 50)$ and $(20, 40)$;
Maximum value of $Z = 400$ at $x = 0, y = 200$.

Question 9

A manufacturer considers that men and women workers are equally efficient and so he pays them at the same rate. He has 30 and 17 units of workers (male and female) and capital respectively, which he uses to produce two types of goods A and B. To produce one unit of A, 2 workers and 3 units of capital are required while 3 workers and 1 unit of capital is required to produce one unit of B. If A and B are priced at Rs. 100 and Rs. 120 per unit respectively, how should he use his resources to maximise the total revenue? Form the above as an L.P.P. and solve graphically.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)

★ Likely Exam Question



Given: Let x be units of good A and y be units of good B manufactured. Worker limit = 30, Capital limit = 17. Price of A = Rs. 100, Price of B = Rs. 120.

To Find: Formulate LPP and maximize revenue Z .

Concept / Formula Used: Objective function: $Z = 100x + 120y$, subject to resource and non-negativity constraints.

Solution:

Let x be the number of units of good A and y be the number of units of good B.

Resource allocation table:

Resource	Good A (x)	Good B (y)	Maximum Availability
Workers	2	3	30
Capital	3	1	17
Profit / Price	Rs. 100	Rs. 120	

Formulate the LPP:

$$\text{Maximise } Z = 100x + 120y$$

Subject to constraints:

$$\begin{aligned} 2x + 3y &\leq 30 \\ 3x + y &\leq 17 \\ x &\geq 0, \quad y \geq 0 \end{aligned}$$

Convert constraints to boundary lines:

$$\begin{aligned} L_1 : 2x + 3y &= 30 \\ L_2 : 3x + y &= 17 \end{aligned}$$

Find intersection of L_1 and L_2 : Multiply L_2 by 3:

$$9x + 3y = 51$$

Subtract L_1 from this equation:

$$\begin{aligned} (9x + 3y) - (2x + 3y) &= 51 - 30 \\ 7x &= 21 \implies x = 3 \end{aligned}$$

Substitute $x = 3$ into L_2 :

$$3(3) + y = 17 \implies 9 + y = 17 \implies y = 8 \implies B(3, 8)$$

Axis intercepts: For L_1 : (15, 0) and (0, 10). For L_2 : $(\frac{17}{3}, 0)$ and (0, 17).

Feasible region corner points are:

$$O(0, 0), \quad A\left(\frac{17}{3}, 0\right), \quad B(3, 8), \quad C(0, 10)$$

Evaluate $Z = 100x + 120y$ at each vertex:

$$\begin{aligned} \text{At } O(0, 0) : \quad Z &= 0 \\ \text{At } A\left(\frac{17}{3}, 0\right) : \quad Z &= 100\left(\frac{17}{3}\right) + 0 = \frac{1700}{3} \approx 566.67 \\ \text{At } B(3, 8) : \quad Z &= 100(3) + 120(8) = 300 + 960 = 1260 \\ \text{At } C(0, 10) : \quad Z &= 100(0) + 120(10) = 1200 \end{aligned}$$

The maximum revenue is Rs. 1260 at $x = 3$ and $y = 8$.

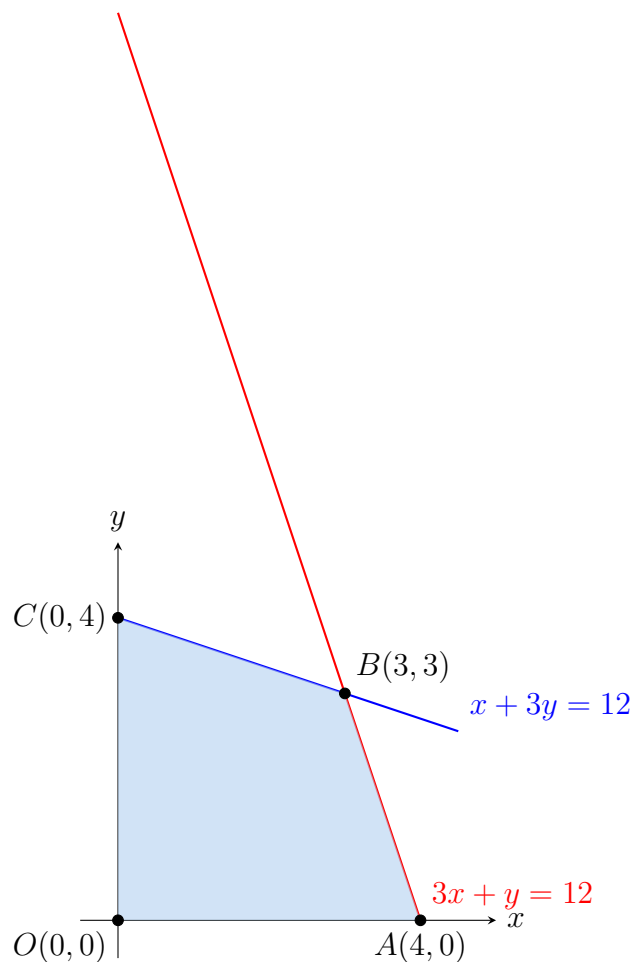
Final Answer

Maximum revenue is Rs. 1260 obtained by producing 3 units of good A and 8 units of good B.

Question 10

A manufacturer produces nuts and bolts. It takes 1 hour of work on machine A and 3 hours on machine B to produce a package of nuts. It takes 3 hours on machine A and 1 hour on machine B to produce a package of bolts. He earns a profit of Rs. 17.50 per package on nuts and Rs. 7 per package on bolts. How many packages of each should be produced each day, so as to maximize his profit, if he operates his machines for at most 12 hours a day?

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)



Feasible region for Question 10

Given: Let x be packages of nuts and y be packages of bolts. Max operating hours per machine = 12. Profit on nuts = Rs. 17.50, Profit on bolts = Rs. 7.00.

To Find: The number of packages of nuts and bolts to maximize profit.

Concept / Formula Used: Objective function: $Z = 17.50x + 7.00y$

Solution:

Let x be the number of packages of nuts and y be the number of packages of bolts.

Constraints setup:

$$\text{Machine A time: } x + 3y \leq 12$$

$$\text{Machine B time: } 3x + y \leq 12$$

$$x \geq 0, \quad y \geq 0$$

Objective function:

$$\text{Maximise } Z = 17.5x + 7y$$

Convert to boundary equations:

$$L_1 : x + 3y = 12$$

$$L_2 : 3x + y = 12$$

Find intersection of L_1 and L_2 : Add L_1 and L_2 :

$$4x + 4y = 24 \implies x + y = 6$$

Subtract L_1 from L_2 :

$$2x - 2y = 0 \implies x = y$$

Thus, $x = 3, y = 3 \implies B(3, 3)$.

Axis intercepts: For L_1 : (12, 0) and (0, 4). For L_2 : (4, 0) and (0, 12).

Feasible region corner points are:

$$O(0, 0), \quad A(4, 0), \quad B(3, 3), \quad C(0, 4)$$

Evaluate $Z = 17.5x + 7y$ at corner points:

$$\text{At } O(0, 0) : Z = 0$$

$$\text{At } A(4, 0) : Z = 17.5(4) + 7(0) = 70$$

$$\text{At } B(3, 3) : Z = 17.5(3) + 7(3) = 52.5 + 21 = 73.5$$

$$\text{At } C(0, 4) : Z = 17.5(0) + 7(4) = 28$$

Maximum profit is Rs. 73.50 obtained when 3 packages of nuts and 3 packages of bolts are produced.

Final Answer

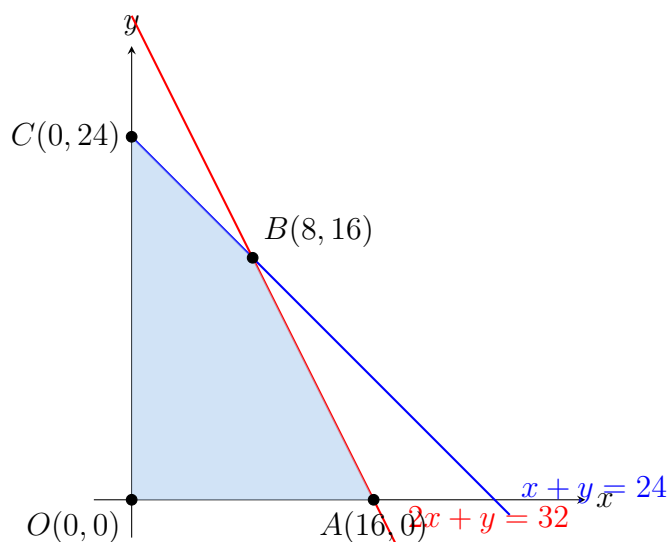
Maximum profit = Rs. 73.50 by producing 3 packages of nuts and 3 packages of bolts.

Question 11

A small firm manufactures gold rings and chains. The combined number of rings and chains manufactured per day is at most 24. It takes one hour to make a ring and half an hour for a chain. The maximum number of hours available per day is 16. If the

profit on a ring is Rs. 300 and on a chain is Rs. 190, how many of each should be manufactured daily so as to maximize the profit?

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)



Feasible region for Question 11

Given: Let x be number of gold rings and y be number of chains. Max combined items = 24. Max hours available = 16. Profit on ring = Rs. 300, Profit on chain = Rs. 190.

To Find: The number of gold rings and chains to maximize total profit.

Concept / Formula Used: Objective function: $Z = 300x + 190y$

Solution:

Let x be the number of gold rings and y be the number of chains.

Formulate constraints:

$$\text{Total production constraint: } x + y \leq 24$$

$$\text{Time constraint: } x + 0.5y \leq 16 \implies 2x + y \leq 32$$

$$x \geq 0, \quad y \geq 0$$

Objective function to maximize:

$$\text{Maximise } Z = 300x + 190y$$

Boundary equations:

$$L_1 : x + y = 24$$

$$L_2 : 2x + y = 32$$

Intersection of L_1 and L_2 : Subtract L_1 from L_2 :

$$\begin{aligned}(2x + y) - (x + y) &= 32 - 24 \\ x &= 8\end{aligned}$$

Substitute $x = 8$ into L_1 :

$$8 + y = 24 \implies y = 16 \implies B(8, 16)$$

Axis intercepts: For L_1 : $(24, 0)$ and $(0, 24)$. For L_2 : $(16, 0)$ and $(0, 32)$.

Feasible region corner points:

$$O(0, 0), \quad A(16, 0), \quad B(8, 16), \quad C(0, 24)$$

Evaluate $Z = 300x + 190y$ at each corner point:

$$\begin{aligned}\text{At } O(0, 0) : \quad Z &= 0 \\ \text{At } A(16, 0) : \quad Z &= 300(16) + 190(0) = 4800 \\ \text{At } B(8, 16) : \quad Z &= 300(8) + 190(16) = 2400 + 3040 = 5440 \\ \text{At } C(0, 24) : \quad Z &= 300(0) + 190(24) = 4560\end{aligned}$$

Maximum profit is Rs. 5440 obtained by manufacturing 8 gold rings and 16 chains.

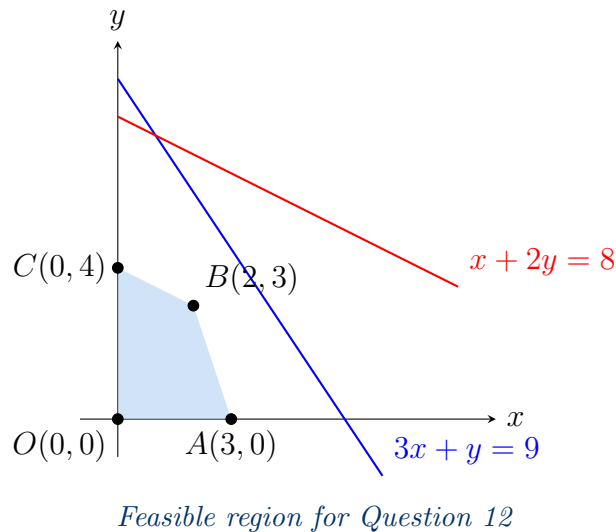
Final Answer

Maximum profit = Rs. 5440 by manufacturing 8 gold rings and 16 chains daily.

Question 12

A company produces two types of goods, A and B, that require gold and silver. Each unit of type A requires 3 g of silver and 1 g of gold while that of B requires 1 g of silver and 2 g of gold. The company can use at most 9 g of silver and 8 g of gold. If each unit of type A brings a profit of Rs. 40 and that of type B brings Rs. 50, find the number of units of each type that the company should produce to maximize the profit. Formulate and solve graphically the LPP and find the maximum profit.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)



Given: Let x be units of good A and y be units of good B. Silver limit = 9 g, Gold limit = 8 g. Profit on A = Rs. 40, Profit on B = Rs. 50.

To Find: Formulate LPP, solve graphically, and determine maximum profit.

Concept / Formula Used: Objective function: $Z = 40x + 50y$

Solution:

Let x be the number of units of type A and y be the number of units of type B.

Formulate constraints:

$$\text{Silver constraint: } 3x + y \leq 9$$

$$\text{Gold constraint: } x + 2y \leq 8$$

$$x \geq 0, \quad y \geq 0$$

Objective function:

$$\text{Maximise } Z = 40x + 50y$$

Boundary equations:

$$L_1 : 3x + y = 9$$

$$L_2 : x + 2y = 8$$

Find intersection of L_1 and L_2 : Multiply L_2 by 3:

$$3x + 6y = 24$$

Subtract L_1 from this equation:

$$\begin{aligned} (3x + 6y) - (3x + y) &= 24 - 9 \\ 5y &= 15 \implies y = 3 \end{aligned}$$

Substitute $y = 3$ into L_2 :

$$x + 2(3) = 8 \implies x = 2 \implies B(2, 3)$$

Axis intercepts: For L_1 : $(3, 0)$ and $(0, 9)$. For L_2 : $(8, 0)$ and $(0, 4)$.

Feasible region corner points:

$$O(0, 0), \quad A(3, 0), \quad B(2, 3), \quad C(0, 4)$$

Evaluate $Z = 40x + 50y$ at each corner point:

$$\text{At } O(0, 0) : Z = 0$$

$$\text{At } A(3, 0) : Z = 40(3) + 50(0) = 120$$

$$\text{At } B(2, 3) : Z = 40(2) + 50(3) = 80 + 150 = 230$$

$$\text{At } C(0, 4) : Z = 40(0) + 50(4) = 200$$

Maximum profit is Rs. 230 by producing 2 units of type A and 3 units of type B.

Final Answer

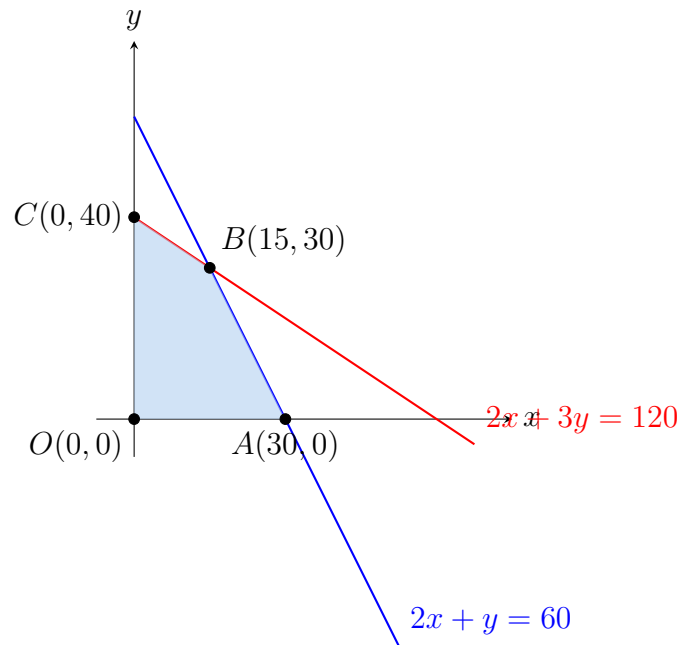
Maximum profit = Rs. 230 obtained by producing 2 units of type A and 3 units of type B.

Question 13

A manufacturer has two machines X and Y that may run at the most 360 minutes in a day to produce two types of toys A and B. To produce each Toy A, machines X and Y need to run at the most 12 minutes and 6 minutes respectively. To produce each Toy B, machines X and Y need to run 6 minutes and 9 minutes respectively. By selling toys A and B, the manufacturer makes the profits of Rs. 30 and Rs. 20 respectively. Formulate a Linear Programming Problem and find the number of toys A and B that should be manufactured in a day to get maximum profit.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)

★ Likely Exam Question



Feasible region for Question 13

Given: Let x be number of Toy A and y be number of Toy B. Max time per machine = 360 mins. Profit on Toy A = Rs. 30, Profit on Toy B = Rs. 20.

To Find: Formulate LPP and maximize daily profit.

Concept / Formula Used: Objective function: $Z = 30x + 20y$

Solution:

Let x be the number of Toy A and y be the number of Toy B manufactured daily.

Constraints derivation (in minutes):

$$\text{Machine X time: } 12x + 6y \leq 360 \implies 2x + y \leq 60$$

$$\text{Machine Y time: } 6x + 9y \leq 360 \implies 2x + 3y \leq 120$$

$$x \geq 0, \quad y \geq 0$$

Objective function:

$$\text{Maximise } Z = 30x + 20y$$

Boundary equations:

$$L_1 : 2x + y = 60$$

$$L_2 : 2x + 3y = 120$$

Intersection of L_1 and L_2 : Subtract L_1 from L_2 :

$$(2x + 3y) - (2x + y) = 120 - 60$$

$$2y = 60 \implies y = 30$$

Substitute $y = 30$ into L_1 :

$$2x + 30 = 60 \implies 2x = 30 \implies x = 15 \implies B(15, 30)$$

Axis intercepts: For L_1 : (30, 0) and (0, 60). For L_2 : (60, 0) and (0, 40).

Feasible region corner points:

$$O(0, 0), \quad A(30, 0), \quad B(15, 30), \quad C(0, 40)$$

Evaluate $Z = 30x + 20y$ at corner points:

$$\text{At } O(0, 0) : Z = 0$$

$$\text{At } A(30, 0) : Z = 30(30) + 20(0) = 900$$

$$\text{At } B(15, 30) : Z = 30(15) + 20(30) = 450 + 600 = 1050$$

$$\text{At } C(0, 40) : Z = 30(0) + 20(40) = 800$$

Maximum profit is Rs. 1050 achieved by producing 15 toys of type A and 30 toys of type B.

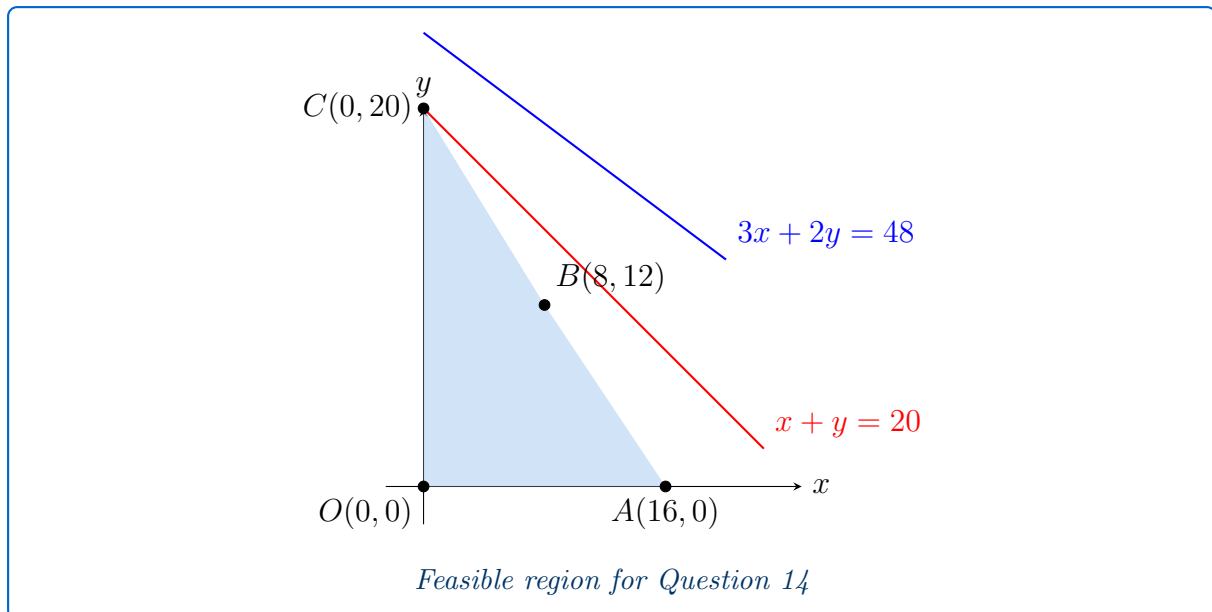
Final Answer

Maximum profit = Rs. 1050 by manufacturing 15 units of Toy A and 30 units of Toy B.

Question 14

A dealer in rural area wishes to purchase a number of sewing machines. He has only Rs. 57600 to invest and has space for at most 20 items. An electronic sewing machine costs him Rs. 3600 and a manually operated sewing machine Rs. 2400. He can sell an electronic sewing machine at a profit of Rs. 220 and a manually operated sewing machine at a profit of Rs. 180. Assuming that he can sell all the items that he can buy, how should he invest his money in order to maximize his profit. Make it as an L.P.P. and solve it graphically.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)



Given: Let x be electronic sewing machines and y be manual sewing machines. Total investment limit = Rs. 57600, Max space limit = 20 items. Cost of electronic = Rs. 3600, Cost of manual = Rs. 2400. Profit on electronic = Rs. 220, Profit on manual = Rs. 180.

To Find: Formulate LPP and solve graphically to maximize profit.

Concept / Formula Used: Objective function: $Z = 220x + 180y$

Solution:

Let x be the number of electronic sewing machines and y be the number of manually operated sewing machines.

Constraints setup:

$$\text{Investment constraint: } 3600x + 2400y \leq 57600 \implies 3x + 2y \leq 48$$

$$\text{Space constraint: } x + y \leq 20$$

$$x \geq 0, \quad y \geq 0$$

Objective function:

$$\text{Maximise } Z = 220x + 180y$$

Boundary lines:

$$L_1 : 3x + 2y = 48$$

$$L_2 : x + y = 20$$

Intersection of L_1 and L_2 : Multiply L_2 by 2:

$$2x + 2y = 40$$

Subtract from L_1 :

$$\begin{aligned}(3x + 2y) - (2x + 2y) &= 48 - 40 \\ x &= 8\end{aligned}$$

Substitute $x = 8$ into L_2 :

$$8 + y = 20 \implies y = 12 \implies B(8, 12)$$

Axis intercepts: For L_1 : (16, 0) and (0, 24). For L_2 : (20, 0) and (0, 20).

Feasible region corner points:

$$O(0, 0), \quad A(16, 0), \quad B(8, 12), \quad C(0, 20)$$

Evaluate $Z = 220x + 180y$ at each corner point:

$$\begin{aligned}\text{At } O(0, 0) : \quad Z &= 0 \\ \text{At } A(16, 0) : \quad Z &= 220(16) + 180(0) = 3520 \\ \text{At } B(8, 12) : \quad Z &= 220(8) + 180(12) = 1760 + 2160 = 3920 \\ \text{At } C(0, 20) : \quad Z &= 220(0) + 180(20) = 3600\end{aligned}$$

Maximum profit is Rs. 3920 obtained by buying 8 electronic sewing machines and 12 manual sewing machines.

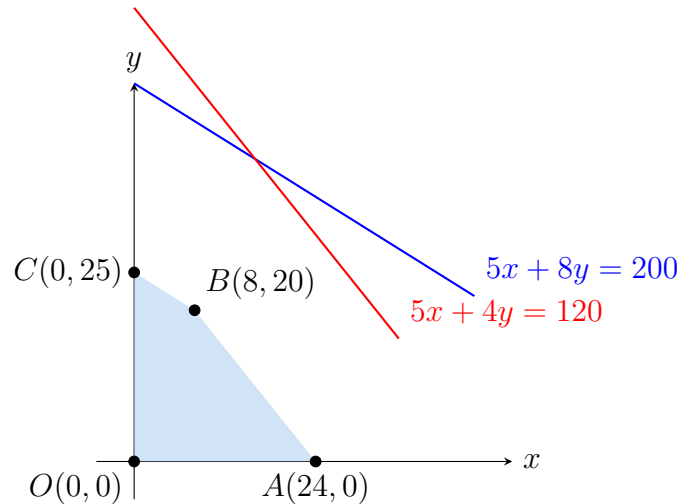
Final Answer

Maximum profit = Rs. 3920 by purchasing 8 electronic and 12 manually operated sewing machines.

Question 15

A company manufactures two types of novelty souvenirs made of plywood. Souvenirs of type A require 5 minutes each for cutting and 10 minutes each for assembling. Souvenirs of type B require 8 minutes each for cutting and 8 minutes each for assembling. There are 3 hours 20 minutes available for cutting and 4 hours available for assembling. The profit is Rs. 50 each for type A and Rs. 60 each for type B souvenirs. How many souvenirs of each type should the company manufacture in order to maximize the profit?

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)



Feasible region for Question 15

Given: Let x be type A souvenirs and y be type B souvenirs. Cutting time available = 3 hours 20 minutes = 200 minutes. Assembling time available = 4 hours = 240 minutes. Profit A = Rs. 50, Profit B = Rs. 60.

To Find: The number of each type of souvenir to maximize profit.

Concept / Formula Used: Objective function: $Z = 50x + 60y$

Solution:

Convert available times into minutes:

$$\text{Cutting time limit} = 3 \text{ hours } 20 \text{ minutes} = 3 \times 60 + 20 = 200 \text{ minutes}$$

$$\text{Assembling time limit} = 4 \text{ hours} = 4 \times 60 = 240 \text{ minutes}$$

Let x be the number of type A souvenirs and y be the number of type B souvenirs.

Formulate constraints:

$$\text{Cutting constraint: } 5x + 8y \leq 200$$

$$\text{Assembling constraint: } 10x + 8y \leq 240 \implies 5x + 4y \leq 120$$

$$x \geq 0, \quad y \geq 0$$

Objective function:

$$\text{Maximise } Z = 50x + 60y$$

Boundary equations:

$$L_1 : 5x + 8y = 200$$

$$L_2 : 5x + 4y = 120$$

Find intersection of L_1 and L_2 : Subtract L_2 from L_1 :

$$\begin{aligned}(5x + 8y) - (5x + 4y) &= 200 - 120 \\ 4y &= 80 \implies y = 20\end{aligned}$$

Substitute $y = 20$ into L_2 :

$$5x + 4(20) = 120 \implies 5x = 40 \implies x = 8 \implies B(8, 20)$$

Axis intercepts: For L_1 : (40, 0) and (0, 25). For L_2 : (24, 0) and (0, 30).

Feasible region corner points:

$$O(0, 0), \quad A(24, 0), \quad B(8, 20), \quad C(0, 25)$$

Evaluate $Z = 50x + 60y$ at each corner point:

$$\begin{aligned}\text{At } O(0, 0) : \quad Z &= 0 \\ \text{At } A(24, 0) : \quad Z &= 50(24) + 60(0) = 1200 \\ \text{At } B(8, 20) : \quad Z &= 50(8) + 60(20) = 400 + 1200 = 1600 \\ \text{At } C(0, 25) : \quad Z &= 50(0) + 60(25) = 1500\end{aligned}$$

Maximum profit is Rs. 1600 obtained by producing 8 souvenirs of type A and 20 souvenirs of type B.

Final Answer

Maximum profit = Rs. 1600 by manufacturing 8 souvenirs of type A and 20 souvenirs of type B.

Question 16

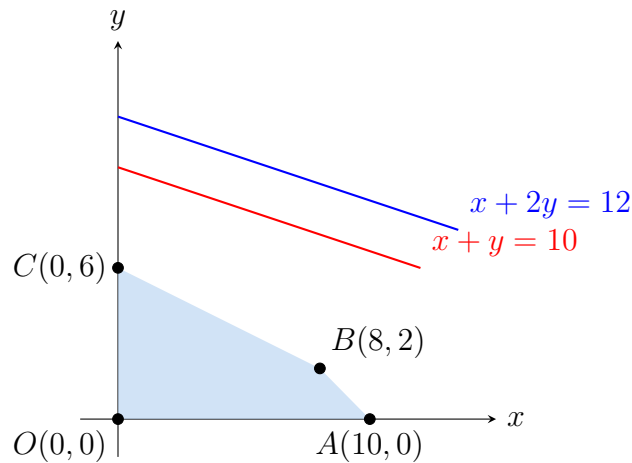
A soap company manufactures two varieties of soap using two machines. The time taken on each machine to manufacture one batch of each variety of soap is given below:

	Machine 1	Machine 2
Soap A	1 hour	1 hour
Soap B	2 hours	1 hour

Machine 1 and Machine 2 cannot be operational for more than 12 and 10 hours in a day, respectively. The profit earned from one batch of Soap A and one batch of Soap B is Rs. 3000 and Rs. 4000, respectively. Find the number of batches of each kind of soap that should be made daily to maximise the profit.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)

★ Likely Exam Question



Feasible region for Question 16

Given: Let x be batches of Soap A and y be batches of Soap B. Machine 1 max = 12 hours, Machine 2 max = 10 hours. Profit A = Rs. 3000, Profit B = Rs. 4000.

To Find: The number of batches of each soap variety to maximize daily profit.

Concept / Formula Used: Objective function: $Z = 3000x + 4000y$

Solution:

Let x be the number of batches of Soap A and y be the number of batches of Soap B.

Formulate constraints:

$$\text{Machine 1 constraint: } x + 2y \leq 12$$

$$\text{Machine 2 constraint: } x + y \leq 10$$

$$x \geq 0, \quad y \geq 0$$

Objective function to maximize:

$$\text{Maximise } Z = 3000x + 4000y$$

Boundary equations:

$$L_1 : x + 2y = 12$$

$$L_2 : x + y = 10$$

Intersection of L_1 and L_2 : Subtract L_2 from L_1 :

$$(x + 2y) - (x + y) = 12 - 10$$

$$y = 2$$

Substitute $y = 2$ into L_2 :

$$x + 2 = 10 \implies x = 8 \implies B(8, 2)$$

Axis intercepts: For L_1 : (12, 0) and (0, 6). For L_2 : (10, 0) and (0, 10).

Feasible region corner points:

$$O(0, 0), \quad A(10, 0), \quad B(8, 2), \quad C(0, 6)$$

Evaluate $Z = 3000x + 4000y$ at corner points:

$$\text{At } O(0, 0) : Z = 0$$

$$\text{At } A(10, 0) : Z = 3000(10) + 4000(0) = 30000$$

$$\text{At } B(8, 2) : Z = 3000(8) + 4000(2) = 24000 + 8000 = 32000$$

$$\text{At } C(0, 6) : Z = 3000(0) + 4000(6) = 24000$$

Maximum profit is Rs. 32000 obtained by making 8 batches of Soap A and 2 batches of Soap B.

Final Answer

Maximum profit = Rs. 32000 by producing 8 batches of Soap A and 2 batches of Soap B.

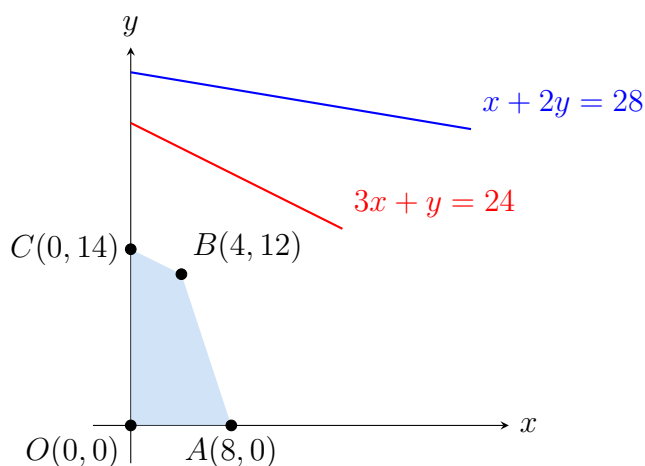
Question 17

A cottage industry manufactures pedestal lamps and wooden shades. Both products require machine time as well as craftsman time in making. The number of hour(s) for producing 1 unit of each and corresponding profit is given in the following table:

Item	Machine time	Craftsman time	Profit (in Rs.)
Pedestal lamp	1.5 hours	3 hours	30
Wooden shade	3 hours	1 hour	20

In a day, the factory has availability of not more than 42 hours of machine time and 24 hours of craftsman time. Assuming that all items manufactured are sold, how should the manufacturer schedule his daily production in order to maximise the profit? Formulate it as an L.P.P. and solve graphically.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)



Feasible region for Question 17

Given: Let x be pedestal lamps and y be wooden shades. Max machine time = 42 hours, Max craftsman time = 24 hours. Profit on lamp = Rs. 30, Profit on shade = Rs. 20.

To Find: Formulate LPP and maximize daily profit.

Concept / Formula Used: Objective function: $Z = 30x + 20y$

Solution:

Let x be the number of pedestal lamps and y be the number of wooden shades.

Formulate constraints:

$$\text{Machine time constraint: } 1.5x + 3y \leq 42 \implies x + 2y \leq 28$$

$$\text{Craftsman time constraint: } 3x + y \leq 24$$

$$x \geq 0, \quad y \geq 0$$

Objective function:

$$\text{Maximise } Z = 30x + 20y$$

Boundary equations:

$$L_1 : x + 2y = 28$$

$$L_2 : 3x + y = 24$$

Find intersection of L_1 and L_2 : Multiply L_2 by 2:

$$6x + 2y = 48$$

Subtract L_1 from this equation:

$$(6x + 2y) - (x + 2y) = 48 - 28$$

$$5x = 20 \implies x = 4$$

Substitute $x = 4$ into L_2 :

$$3(4) + y = 24 \implies 12 + y = 24 \implies y = 12 \implies B(4, 12)$$

Axis intercepts: For L_1 : (28, 0) and (0, 14). For L_2 : (8, 0) and (0, 24).

Feasible region corner points:

$$O(0, 0), \quad A(8, 0), \quad B(4, 12), \quad C(0, 14)$$

Evaluate $Z = 30x + 20y$ at corner points:

$$\text{At } O(0, 0) : Z = 0$$

$$\text{At } A(8, 0) : Z = 30(8) + 20(0) = 240$$

$$\text{At } B(4, 12) : Z = 30(4) + 20(12) = 120 + 240 = 360$$

$$\text{At } C(0, 14) : Z = 30(0) + 20(14) = 280$$

Maximum profit is Rs. 360 obtained by producing 4 pedestal lamps and 12 wooden shades.

Final Answer

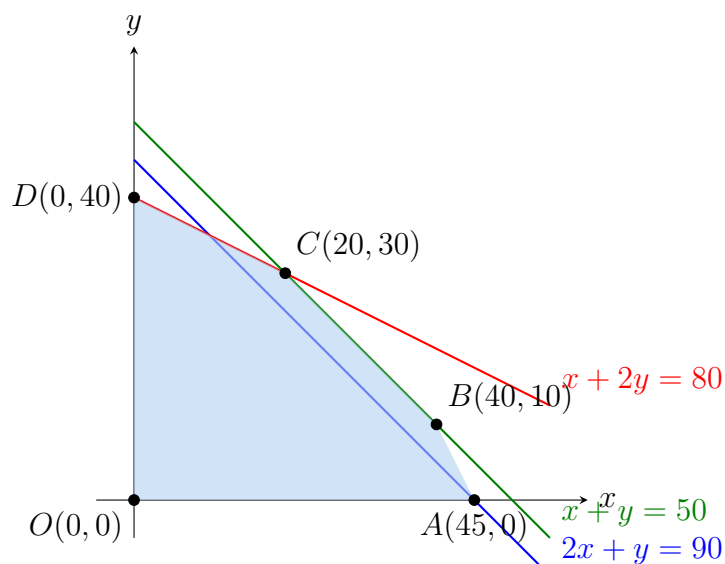
Maximum profit = Rs. 360 by producing 4 pedestal lamps and 12 wooden shades.

Question 18

A carpenter has 90, 80 and 50 running feet respectively of teak wood, plywood and rosewood which is used to produce product A and product B. Each unit of product A requires 2, 1 and 1 running feet and each unit of product B requires 1, 2 and 1 running feet of teak wood, plywood and rosewood respectively. If product A is sold for Rs. 48 per unit and product B is sold for Rs. 40 per unit, how many units of product A and product B should be produced and sold by the carpenter, in order to obtain the maximum revenue. Formulate the above as a linear programming problem and solve it, indicating clearly the feasible region in the graph.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)

★ Likely Exam Question



Feasible region for Question 18

Given: Let x be units of product A and y be units of product B. Teak wood limit = 90 ft, Plywood limit = 80 ft, Rosewood limit = 50 ft. Selling price of A = Rs. 48, Selling price of B = Rs. 40.

To Find: Formulate LPP and determine units of A and B to maximize total revenue.

Concept / Formula Used: Objective function: $Z = 48x + 40y$

Solution:

Let x be the number of units of product A and y be the number of units of product B.

Formulate constraints:

$$\text{Teak wood constraint: } 2x + y \leq 90$$

$$\text{Plywood constraint: } x + 2y \leq 80$$

$$\text{Rosewood constraint: } x + y \leq 50$$

$$x \geq 0, \quad y \geq 0$$

Objective function:

$$\text{Maximise } Z = 48x + 40y$$

Boundary lines:

$$L_1 : 2x + y = 90$$

$$L_2 : x + 2y = 80$$

$$L_3 : x + y = 50$$

Find relevant boundary intersection points: 1. Intersection of L_1 and L_3 : Subtract L_3 from L_1 :

$$\begin{aligned} (2x + y) - (x + y) &= 90 - 50 \\ x &= 40 \end{aligned}$$

Substitute $x = 40$ into L_3 :

$$40 + y = 50 \implies y = 10 \implies B(40, 10)$$

Check in L_2 : $40 + 2(10) = 60 \leq 80$ (Satisfied).

2. Intersection of L_2 and L_3 : Subtract L_3 from L_2 :

$$\begin{aligned} (x + 2y) - (x + y) &= 80 - 50 \\ y &= 30 \end{aligned}$$

Substitute $y = 30$ into L_3 :

$$x + 30 = 50 \implies x = 20 \implies C(20, 30)$$

Check in L_1 : $2(20) + 30 = 70 \leq 90$ (Satisfied).

Axis intercepts on boundaries: For L_1 on $y = 0$: $x = 45 \implies A(45, 0)$ (Satisfies $45 + 0 \leq 80$ and $45 + 0 \leq 50$). For L_2 on $x = 0$: $y = 40 \implies D(0, 40)$ (Satisfies $0 + 40 \leq 90$ and $0 + 40 \leq 50$).

The corner points of the bounded feasible polygon are:

$$O(0, 0), \quad A(45, 0), \quad B(40, 10), \quad C(20, 30), \quad D(0, 40)$$

Evaluate $Z = 48x + 40y$ at each corner point:

$$\text{At } O(0, 0) : Z = 0$$

$$\text{At } A(45, 0) : Z = 48(45) + 40(0) = 2160$$

$$\text{At } B(40, 10) : Z = 48(40) + 40(10) = 1920 + 400 = 2320$$

$$\text{At } C(20, 30) : Z = 48(20) + 40(30) = 960 + 1200 = 2160$$

$$\text{At } D(0, 40) : Z = 48(0) + 40(40) = 1600$$

Maximum revenue is Rs. 2320 obtained by producing 40 units of product A and 10 units of product B.

Final Answer

Maximum revenue = Rs. 2320 obtained by producing 40 units of product A and 10 units of product B.

Question 19

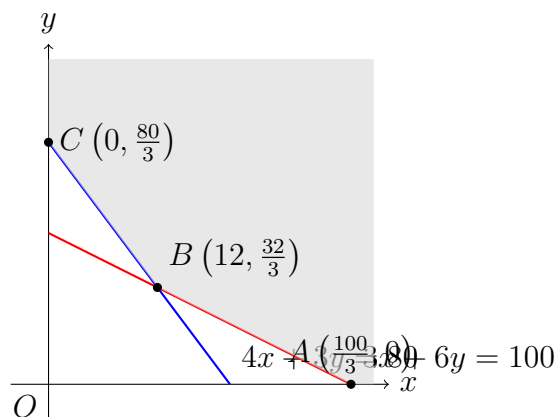
A diet is to contain at least 80 units of Vitamin A and 100 units of minerals. Two foods F_1 and F_2 are available costing Rs. 5 per unit and Rs. 6 per unit respectively. One unit of food F_1 contains 4 units of vitamin A and 3 units of minerals whereas one unit of food F_2 contains 3 units of vitamin A and 6 units of minerals. Formulate this as a linear programming problem. Find the minimum cost of diet that consists of mixture of these two foods and also meets minimum nutritional requirement.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)

★ Likely Exam Question

Given: Diet requirements: Vitamin A ≥ 80 units, Minerals ≥ 100 units. Cost of F_1 = Rs. 5/unit, Cost of F_2 = Rs. 6/unit. Food F_1 provides 4 units Vitamin A, 3 units Minerals. Food F_2 provides 3 units Vitamin A, 6 units Minerals.

Concept / Formula Used: Objective function: $Z = c_1x + c_2y$, subject to nutritional constraints $a_ix + b_iy \geq d_i$ and non-negativity $x \geq 0, y \geq 0$.



Feasible region for Question 19

Solution:

Let x units of food F_1 and y units of food F_2 be included in the diet.

Since quantities cannot be negative, we have:

$$x \geq 0, \quad y \geq 0$$

The total cost Z (in Rs.) to be minimized is:

$$Z = 5x + 6y$$

The nutritional requirements give the following linear constraints: 1) Vitamin A constraint:

$$4x + 3y \geq 80$$

2) Minerals constraint:

$$3x + 6y \geq 100$$

To determine the feasible region, we convert inequalities to equations: Line $L_1 : 4x + 3y = 80$ passes through $(20, 0)$ and $(0, \frac{80}{3})$. Line $L_2 : 3x + 6y = 100$ passes through $(\frac{100}{3}, 0)$ and $(0, \frac{50}{3})$.

We find the point of intersection B of L_1 and L_2 by solving them simultaneously: Multiply L_1 by 2:

$$8x + 6y = 160$$

Subtract L_2 ($3x + 6y = 100$) from this equation:

$$(8x + 6y) - (3x + 6y) = 160 - 100$$

$$5x = 60$$

$$x = 12$$

Substitute $x = 12$ into L_1 :

$$4(12) + 3y = 80$$

$$48 + 3y = 80$$

$$3y = 32$$

$$y = \frac{32}{3}$$

Thus, point $B = (12, \frac{32}{3})$.

The corner points of the unbounded feasible region are:

$$A\left(\frac{100}{3}, 0\right), \quad B\left(12, \frac{32}{3}\right), \quad C\left(0, \frac{80}{3}\right)$$

Evaluating the objective function $Z = 5x + 6y$ at each corner point: At $A\left(\frac{100}{3}, 0\right)$:

$$\begin{aligned} Z &= 5\left(\frac{100}{3}\right) + 6(0) \\ &= \frac{500}{3} \approx 166.67 \end{aligned}$$

At $B\left(12, \frac{32}{3}\right)$:

$$\begin{aligned} Z &= 5(12) + 6\left(\frac{32}{3}\right) \\ &= 60 + 64 \\ &= 124 \end{aligned}$$

At $C\left(0, \frac{80}{3}\right)$:

$$\begin{aligned} Z &= 5(0) + 6\left(\frac{80}{3}\right) \\ &= 160 \end{aligned}$$

Since the feasible region is unbounded, we check if $5x + 6y < 124$ has any common points with the feasible region. The open half-plane $5x + 6y < 124$ has no common points with the feasible region. Therefore, the minimum value of Z is 124.

Final Answer

Minimum cost of the diet is Rs. 124, achieved by using 12 units of food F_1 and $\frac{32}{3}$ units of food F_2 .

Common Mistake: Since the feasible region is unbounded, failing to verify that the open half-plane $5x + 6y < 124$ shares no common points with the feasible region can cause loss of marks.

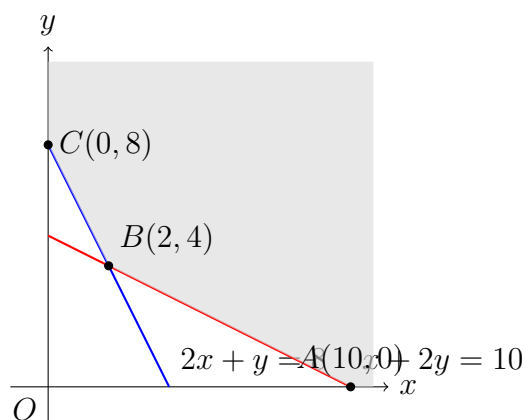
Question 20

A dietician wishes to mix two types of food in such a way that the vitamin contents of the mixture contain at least 8 units of vitamin A, and 10 units of vitamin C. Food I contains 2 units/kg of vitamin A and 1 unit/kg of vitamin C. Food II contains 1 unit/kg of vitamin A and 2 units/kg of vitamin C. It costs Rs. 50 per kg to purchase food I and Rs. 70 per kg to purchase food II. Determine the minimum cost of such a mixture.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)

Given: Vitamin A ≥ 8 units, Vitamin C ≥ 10 units. Food I: 2 units/kg Vitamin A, 1 unit/kg Vitamin C, Cost = Rs. 50/kg. Food II: 1 unit/kg Vitamin A, 2 units/kg Vitamin C, Cost = Rs. 70/kg.

Concept / Formula Used: Objective function: $Z = ax + by$



Feasible region for Question 20

Solution:

Let x kg of Food I and y kg of Food II be mixed.

Non-negativity constraints:

$$x \geq 0, \quad y \geq 0$$

Objective function to minimize:

$$Z = 50x + 70y$$

Subject to constraints: 1) Vitamin A constraint:

$$2x + y \geq 8$$

2) Vitamin C constraint:

$$x + 2y \geq 10$$

Convert inequalities to linear equations: Line $L_1 : 2x + y = 8 \implies$ passes through $(0, 8)$ and $(4, 0)$
 Line $L_2 : x + 2y = 10 \implies$ passes through $(0, 5)$ and $(10, 0)$

Solving $2x + y = 8$ and $x + 2y = 10$ simultaneously for intersection point B : From L_1 :

$$y = 8 - 2x$$

Substitute y into L_2 :

$$\begin{aligned}x + 2(8 - 2x) &= 10 \\x + 16 - 4x &= 10 \\-3x &= -6 \\x &= 2\end{aligned}$$

Substitute $x = 2$ back to find y :

$$\begin{aligned}y &= 8 - 2(2) \\&= 4\end{aligned}$$

Intersection point is $B(2, 4)$.

The corner points of the unbounded feasible region are:

$$A(10, 0), \quad B(2, 4), \quad C(0, 8)$$

Evaluating $Z = 50x + 70y$ at each corner point: At $A(10, 0)$:

$$\begin{aligned}Z &= 50(10) + 70(0) \\&= 500\end{aligned}$$

At $B(2, 4)$:

$$\begin{aligned}Z &= 50(2) + 70(4) \\&= 100 + 280 \\&= 380\end{aligned}$$

At $C(0, 8)$:

$$\begin{aligned}Z &= 50(0) + 70(8) \\&= 560\end{aligned}$$

The open half-plane $50x + 70y < 380 \implies 5x + 7y < 38$ has no common points with the feasible region.

Final Answer

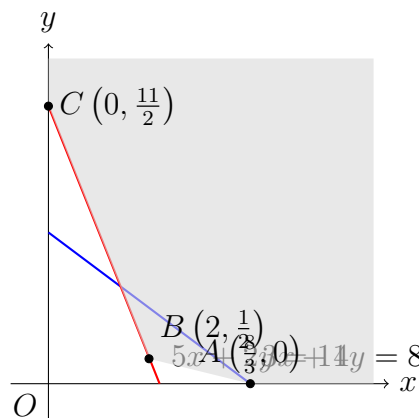
Minimum cost of the mixture is Rs. 380, obtained by mixing 2 kg of Food I and 4 kg of Food II.

Question 21

Reshma wishes to mix two types of foods P and Q in such a way that the vitamin contents of the mixture contains at least 8 units of vitamin A and 11 units of vitamin B. Food P costs Rs. 60/kg and food Q costs Rs. 80/kg. Food P contains 3 units/kg of vitamin A and 5 units/kg of vitamin B while food Q contains 4 units/kg of vitamin A and 2 units/kg of vitamin B. Determine the minimum cost of the mixture.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)

Given: Food P: Cost = Rs. 60/kg, Vitamin A = 3 units/kg, Vitamin B = 5 units/kg.
Food Q: Cost = Rs. 80/kg, Vitamin A = 4 units/kg, Vitamin B = 2 units/kg. Minimum requirements: Vitamin A ≥ 8 , Vitamin B ≥ 11 .



Feasible region for Question 21

Solution:

Let x kg of Food P and y kg of Food Q be mixed in the mixture.

Constraints:

$$x \geq 0, \quad y \geq 0$$

$$3x + 4y \geq 8 \quad (\text{Vitamin A constraint})$$

$$5x + 2y \geq 11 \quad (\text{Vitamin B constraint})$$

Objective function to minimize:

$$Z = 60x + 80y$$

To find intersection point B of $3x + 4y = 8$ and $5x + 2y = 11$: Multiply $5x + 2y = 11$ by 2:

$$10x + 4y = 22$$

Subtract $3x + 4y = 8$:

$$(10x + 4y) - (3x + 4y) = 22 - 8$$

$$7x = 14$$

$$x = 2$$

Substitute $x = 2$ into $3x + 4y = 8$:

$$3(2) + 4y = 8$$

$$6 + 4y = 8$$

$$4y = 2$$

$$y = \frac{1}{2}$$

Corner points of the unbounded feasible region:

$$A\left(\frac{8}{3}, 0\right), \quad B\left(2, \frac{1}{2}\right), \quad C\left(0, \frac{11}{2}\right)$$

Evaluating $Z = 60x + 80y$ at each corner point: At $A\left(\frac{8}{3}, 0\right)$:

$$\begin{aligned} Z &= 60\left(\frac{8}{3}\right) + 80(0) \\ &= 160 \end{aligned}$$

At $B\left(2, \frac{1}{2}\right)$:

$$\begin{aligned} Z &= 60(2) + 80\left(\frac{1}{2}\right) \\ &= 120 + 40 \\ &= 160 \end{aligned}$$

At $C\left(0, \frac{11}{2}\right)$:

$$\begin{aligned} Z &= 60(0) + 80\left(\frac{11}{2}\right) \\ &= 440 \end{aligned}$$

The minimum cost is Rs. 160, which occurs at both corner points $A\left(\frac{8}{3}, 0\right)$ and $B\left(2, \frac{1}{2}\right)$.

Final Answer

Minimum cost is Rs. 160 at all points on the line segment joining $\left(\frac{8}{3}, 0\right)$ and $\left(2, \frac{1}{2}\right)$.

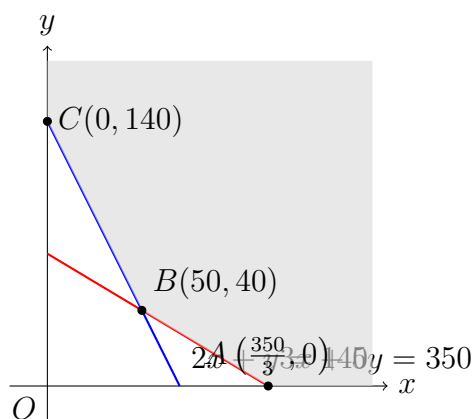
Question 22

A farmer has a supply of chemical fertiliser of type A which contains 10% nitrogen and 6% phosphoric acid and of type B which contains 5% nitrogen and 10% phosphoric acid. After soil test, it is found that at least 7 kg of nitrogen and same quantity of phosphoric acid is required for a good crop. The fertiliser of type A costs Rs. 5 per kg and the type B cost Rs. 8 per kg. Using linear programming, find how many kilograms of each type of the fertiliser should be bought to meet the requirement and for the cost to be minimum. Find the feasible region in the graph.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)

Given: Fertiliser A: 10% Nitrogen, 6% Phosphoric acid, Cost = Rs. 5/kg. Fertiliser B: 5% Nitrogen, 10% Phosphoric acid, Cost = Rs. 8/kg. Requirement: Nitrogen ≥ 7 kg, Phosphoric acid ≥ 7 kg.

Concept / Formula Used: Percentage amount calculation: $p\%$ of $x = \frac{p}{100}x$



Feasible region for Question 22

Solution:

Let x kg of fertiliser A and y kg of fertiliser B be bought.

Non-negativity constraints:

$$x \geq 0, \quad y \geq 0$$

Nitrogen constraint:

$$\begin{aligned} \frac{10}{100}x + \frac{5}{100}y &\geq 7 \\ 10x + 5y &\geq 700 \\ 2x + y &\geq 140 \end{aligned}$$

Phosphoric acid constraint:

$$\begin{aligned} \frac{6}{100}x + \frac{10}{100}y &\geq 7 \\ 6x + 10y &\geq 700 \\ 3x + 5y &\geq 350 \end{aligned}$$

Objective function to minimize:

$$Z = 5x + 8y$$

Solving $2x + y = 140$ and $3x + 5y = 350$ simultaneously: From $2x + y = 140$:

$$y = 140 - 2x$$

Substitute into $3x + 5y = 350$:

$$\begin{aligned}3x + 5(140 - 2x) &= 350 \\3x + 700 - 10x &= 350 \\-7x &= -350 \\x &= 50\end{aligned}$$

Substitute $x = 50$ to find y :

$$\begin{aligned}y &= 140 - 2(50) \\&= 40\end{aligned}$$

Intersection point is $B(50, 40)$.

Corner points of unbounded feasible region:

$$A\left(\frac{350}{3}, 0\right), \quad B(50, 40), \quad C(0, 140)$$

Evaluating $Z = 5x + 8y$ at each corner point: At $A\left(\frac{350}{3}, 0\right)$:

$$\begin{aligned}Z &= 5\left(\frac{350}{3}\right) + 8(0) \\&= \frac{1750}{3} \approx 583.33\end{aligned}$$

At $B(50, 40)$:

$$\begin{aligned}Z &= 5(50) + 8(40) \\&= 250 + 320 \\&= 570\end{aligned}$$

At $C(0, 140)$:

$$\begin{aligned}Z &= 5(0) + 8(140) \\&= 1120\end{aligned}$$

The open half-plane $5x + 8y < 570$ does not overlap with the feasible region.

Final Answer

The farmer should buy 50 kg of fertiliser A and 40 kg of fertiliser B. Minimum cost is Rs. 570.

Question 23

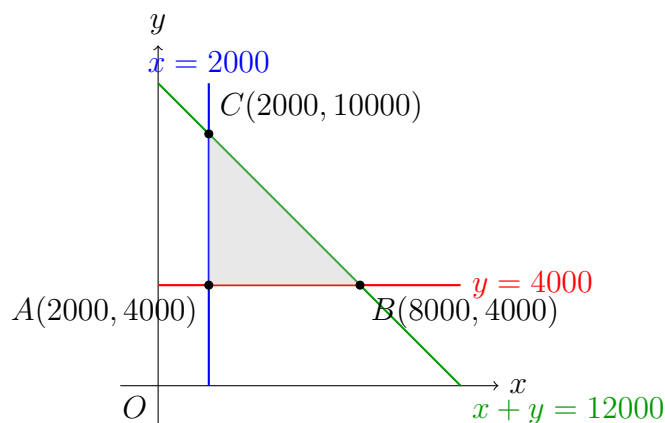
David wants to invest at most Rs. 12000 in Bonds A and B. According to the rule, he has to invest at least Rs. 2000 in Bond A and at least Rs. 4000 in Bond B. If the rates of interest on Bonds A and B respectively are 8% and 10% per annum, formulate the

problem as L.P.P. and solve it graphically for maximum interest. Also determine the maximum interest received in a year.

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)

Given: Total investment $\leq$ Rs. 12000. Investment in Bond A $x \geq 2000$, Investment in Bond B $y \geq 4000$. Interest rate A = 8%, Interest rate B = 10%.

Concept / Formula Used: Interest = Principal $\times$ Rate. Objective function: $Z = 0.08x + 0.10y$



Feasible region for Question 23

Solution:

Let x be the amount invested in Bond A (in Rs.) and y be the amount invested in Bond B (in Rs.).

Constraints:

$$x \geq 2000$$

$$y \geq 4000$$

$$x + y \leq 12000$$

Objective function to maximize annual interest Z :

$$\begin{aligned} Z &= \frac{8}{100}x + \frac{10}{100}y \\ &= 0.08x + 0.10y \end{aligned}$$

The boundary lines intersect at three corner points: 1) Point A: Intersection of $x = 2000$ and $y = 4000 \Rightarrow A(2000, 4000)$. 2) Point B: Intersection of $y = 4000$ and $x + y = 12000$:

$$x + 4000 = 12000$$

$$x = 8000 \Rightarrow B(8000, 4000)$$

3) Point C : Intersection of $x = 2000$ and $x + y = 12000$:

$$\begin{aligned} 2000 + y &= 12000 \\ y &= 10000 \implies C(2000, 10000) \end{aligned}$$

Evaluating $Z = 0.08x + 0.10y$ at each vertex: At $A(2000, 4000)$:

$$\begin{aligned} Z &= 0.08(2000) + 0.10(4000) \\ &= 160 + 400 \\ &= 560 \end{aligned}$$

At $B(8000, 4000)$:

$$\begin{aligned} Z &= 0.08(8000) + 0.10(4000) \\ &= 640 + 400 \\ &= 1040 \end{aligned}$$

At $C(2000, 10000)$:

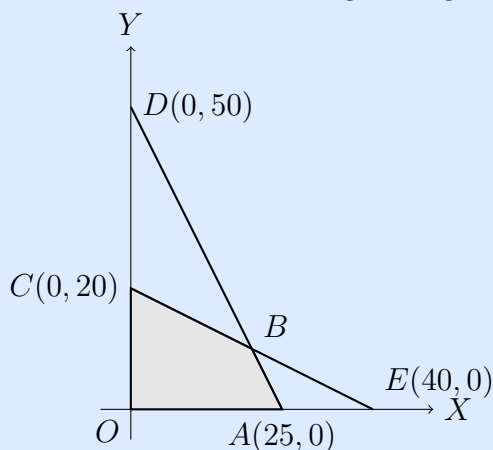
$$\begin{aligned} Z &= 0.08(2000) + 0.10(10000) \\ &= 160 + 1000 \\ &= 1160 \end{aligned}$$

Final Answer

Maximum interest received in a year is Rs. 1160, obtained by investing Rs. 2000 in Bond A and Rs. 10000 in Bond B.

Question 24

The feasible region for an L.P.P. is shown in the given figure:



Based on the given graph, answer the following questions:

- Write the constraints for the L.P.P.
- Find the coordinates of the point B .
- Find the maximum value of the objective function $Z = x + y$.

TYPE 6 Feasible Region Analysis – Reverse-Engineering Constraints from a Graph

Given: Feasible region vertices $O(0, 0)$, $A(25, 0)$, B , $C(0, 20)$. Boundary line through $D(0, 50)$ and $A(25, 0)$, boundary line through $C(0, 20)$ and $E(40, 0)$.

Concept / Formula Used: Intercept form of a line: $\frac{x}{a} + \frac{y}{b} = 1$

Solution:

(a) Equation of line passing through $A(25, 0)$ and $D(0, 50)$:

$$\begin{aligned}\frac{x}{25} + \frac{y}{50} &= 1 \\ 2x + y &= 50\end{aligned}$$

Since the shaded region lies towards the origin from this line, the constraint is:

$$2x + y \leq 50$$

Equation of line passing through $C(0, 20)$ and $E(40, 0)$:

$$\begin{aligned}\frac{x}{40} + \frac{y}{20} &= 1 \\ x + 2y &= 40\end{aligned}$$

Since the shaded region lies towards the origin from this line, the constraint is:

$$x + 2y \leq 40$$

Non-negativity conditions:

$$x \geq 0, \quad y \geq 0$$

(b) Point B is the intersection of $2x + y = 50$ and $x + 2y = 40$. Multiply second equation by 2:

$$2x + 4y = 80$$

Subtract first equation $2x + y = 50$:

$$\begin{aligned}(2x + 4y) - (2x + y) &= 80 - 50 \\ 3y &= 30 \\ y &= 10\end{aligned}$$

Substitute $y = 10$ into $2x + y = 50$:

$$\begin{aligned}2x + 10 &= 50 \\ 2x &= 40 \\ x &= 20\end{aligned}$$

Therefore, $B = (20, 10)$.

(c) The vertices of the feasible region are $O(0, 0)$, $A(25, 0)$, $B(20, 10)$, $C(0, 20)$. Evaluating $Z = x + y$ at each vertex: At $O(0, 0)$: $Z = 0 + 0 = 0$ At $A(25, 0)$: $Z = 25 + 0 = 25$ At $B(20, 10)$: $Z = 20 + 10 = 30$ At $C(0, 20)$: $Z = 0 + 20 = 20$

Final Answer

- (a) Constraints: $2x + y \leq 50$, $x + 2y \leq 40$, $x \geq 0$, $y \geq 0$
 (b) Coordinates of point B : $(20, 10)$
 (c) Maximum value of $Z = 30$ at point $B(20, 10)$

Question 25

An oil company has two depots A and B with capacities of 7000 litre and 4000 litre respectively. The company is to supply oil to three petrol pumps D, E and F, whose requirements are 4500 litre, 3000 litre and 3500 litre respectively. The distance (in km) between the depots and the petrol pumps is given in the following table:

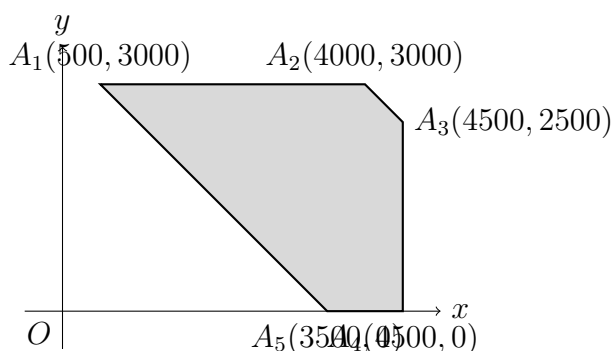
From / To	A	B
D	7	3
E	6	4
F	3	2

Assuming that the transportation cost per km is Rs. 1 per litre, how should the delivery be scheduled in order that the transportation cost is minimum? What is the minimum cost?

TYPE 5 LPP Formulation and Graphical Solution (Full Word Problem)

Given: Capacities: $A = 7000$ L, $B = 4000$ L. Requirements: $D = 4500$ L, $E = 3000$ L, $F = 3500$ L. Distances as in table. Cost = Rs. $1/\text{L} \cdot \text{km}$.

Concept / Formula Used: Total cost $Z = \sum(\text{Quantity}) \times (\text{Distance}) \times (\text{Rate})$



Feasible region pentagon for Question 25

Solution:

Let Depot A supply x litres to petrol pump D, and y litres to petrol pump E.

Since Depot A capacity is 7000 litres:

$$\text{Supply from A to F} = 7000 - x - y$$

Since requirement at D is 4500 litres:

$$\text{Supply from B to D} = 4500 - x$$

Since requirement at E is 3000 litres:

$$\text{Supply from B to E} = 3000 - y$$

Since Depot B capacity is 4000 litres:

$$\begin{aligned}\text{Supply from B to F} &= 4000 - (4500 - x + 3000 - y) \\ &= x + y - 3500\end{aligned}$$

Non-negativity constraints:

$$\begin{aligned}x &\geq 0, \quad y \geq 0 \\ 4500 - x &\geq 0 \implies x \leq 4500 \\ 3000 - y &\geq 0 \implies y \leq 3000 \\ 7000 - x - y &\geq 0 \implies x + y \leq 7000 \\ x + y - 3500 &\geq 0 \implies x + y \geq 3500\end{aligned}$$

Transportation cost function Z :

$$\begin{aligned}Z &= 7x + 6y + 3(7000 - x - y) + 3(4500 - x) + 4(3000 - y) + 2(x + y - 3500) \\ &= 7x + 6y + 21000 - 3x - 3y + 13500 - 3x + 12000 - 4y + 2x + 2y - 7000 \\ &= (7 - 3 - 3 + 2)x + (6 - 3 - 4 + 2)y + (21000 + 13500 + 12000 - 7000) \\ &= 3x + y + 39500\end{aligned}$$

The feasible region is a pentagon with vertices:

$$\begin{aligned}A_1 &= (500, 3000) \\ A_2 &= (4000, 3000) \\ A_3 &= (4500, 2500) \\ A_4 &= (4500, 0) \\ A_5 &= (3500, 0)\end{aligned}$$

Evaluating $Z = 3x + y + 39500$ at each corner point: At $A_1(500, 3000)$:

$$\begin{aligned}Z &= 3(500) + 3000 + 39500 \\ &= 1500 + 3000 + 39500 \\ &= 44000\end{aligned}$$

At $A_2(4000, 3000)$:

$$\begin{aligned}Z &= 3(4000) + 3000 + 39500 \\ &= 12000 + 3000 + 39500 \\ &= 54500\end{aligned}$$

At $A_3(4500, 2500)$:

$$\begin{aligned} Z &= 3(4500) + 2500 + 39500 \\ &= 13500 + 2500 + 39500 \\ &= 55500 \end{aligned}$$

At $A_4(4500, 0)$:

$$\begin{aligned} Z &= 3(4500) + 0 + 39500 \\ &= 13500 + 39500 \\ &= 53000 \end{aligned}$$

At $A_5(3500, 0)$:

$$\begin{aligned} Z &= 3(3500) + 0 + 39500 \\ &= 10500 + 39500 \\ &= 50000 \end{aligned}$$

Minimum cost is Rs. 44000 at $x = 500, y = 3000$.

Delivery Schedule: From Depot A: 500 L to D, 3000 L to E, 3500 L to F. From Depot B: 4000 L to D, 0 L to E, 0 L to F.

Final Answer

Minimum transportation cost is Rs. 44000. Delivery schedule: Depot A supplies 500 L, 3000 L, 3500 L to D, E, F respectively. Depot B supplies 4000 L, 0 L, 0 L to D, E, F respectively.

Question 26

A linear programming problem is given by $Z = px + qy$ where $p, q > 0$ subject to the constraints:

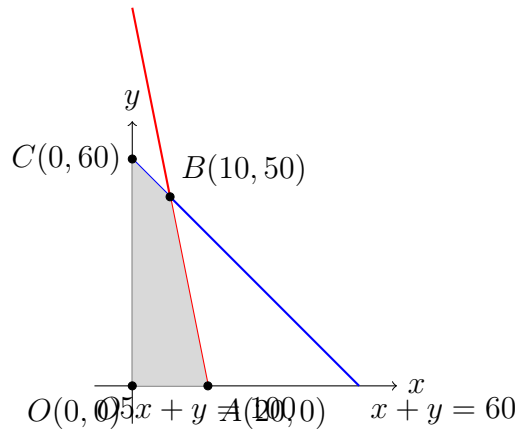
$$x + y \leq 60, \quad 5x + y \leq 100, \quad x \geq 0 \quad \text{and} \quad y \geq 0$$

- Solve graphically to find the corner points of the feasible region.
- If $Z = px + qy$ is maximum at $(0, 60)$ and $(10, 50)$, find the relation of p and q . Also mention the number of optimal solution(s) in this case.

TYPE 4b Corner Point Method – Parameter Relation from Multiple Optima

Given: $Z = px + qy$ with $p, q > 0$. Constraints: $x + y \leq 60, 5x + y \leq 100, x \geq 0, y \geq 0$. Maximum occurs at $(0, 60)$ and $(10, 50)$.

Concept / Formula Used: Equate the objective function $Z = px + qy$ at the two given points P and Q where the maximum occurs ($Z_P = Z_Q$) to determine the parameter relation.



Feasible region for Question 26

Solution:

(i) Lines bounding the region: Line $L_1 : x + y = 60 \implies (0, 60)$ and $(60, 0)$ Line $L_2 : 5x + y = 100 \implies (0, 100)$ and $(20, 0)$

Solving $x + y = 60$ and $5x + y = 100$ simultaneously for intersection point B : Subtract $x + y = 60$ from $5x + y = 100$:

$$\begin{aligned}(5x + y) - (x + y) &= 100 - 60 \\ 4x &= 40 \\ x &= 10\end{aligned}$$

Substitute $x = 10$ into $x + y = 60$:

$$\begin{aligned}10 + y &= 60 \\ y &= 50\end{aligned}$$

So $B = (10, 50)$.

The corner points of the feasible region $OABC$ are:

$$O(0, 0), \quad A(20, 0), \quad B(10, 50), \quad C(0, 60)$$

(ii) Since $Z = px + qy$ is maximum at both $(0, 60)$ and $(10, 50)$, the value of Z must be equal at both points: Value at $(0, 60)$:

$$Z_C = p(0) + q(60) = 60q$$

Value at $(10, 50)$:

$$Z_B = p(10) + q(50) = 10p + 50q$$

Equating Z_C and Z_B :

$$\begin{aligned}10p + 50q &= 60q \\ 10p &= 60q - 50q \\ 10p &= 10q \\ p &= q\end{aligned}$$

Since the maximum value occurs at two adjacent corner points, it occurs at every point on the line segment joining $(0, 60)$ and $(10, 50)$. Thus, there are infinitely many optimal solutions.

Final Answer

- (i) Corner points: $(0, 0), (20, 0), (10, 50), (0, 60)$
- (ii) Relation: $p = q$. There are infinitely many optimal solutions.